

ProteinDelta

User Manual

A guided, illustrated walkthrough of every section of the web application

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1. Introduction

ProteinDelta is a web application that compares the structure of a wild-type protein (WT) with that of its mutated version (MUT) and estimates whether the mutation is pathogenic. It combines AlphaFold2 per-residue confidence (pLDDT), stability ($\Delta\Delta G$), evolutionary fitness, structural displacement (RMSD), change in solvent-accessible surface area ($\Delta SASA$) and fractal dimension (FD) into a single pathogenicity score (PS_v4).

This manual walks through the application from start to finish, section by section, showing where to click and what you get at each step. The tour starts with the application already open: the sign-in screen is not covered because it will be removed in the next version.

1.1 Requirements

- An up-to-date browser (Chrome, Edge, Firefox or Safari) with WebGL enabled, which the 3D viewer needs.
- An internet connection: the application loads its libraries (Chart.js, 3Dmol.js) from a CDN and downloads structures from the Protein Data Bank (RCSB).
- For real-data analyses: a Google account to run the notebook in Google Colab with a GPU.

1.2 How to read the screenshots

All screenshots were taken automatically with Playwright on the real application, with the interface set to English. Two kinds of marks were added on top of them:

- **Numbered red box:** points to an element on the screen. A legend below the figure explains each number.
- **Cursor with a yellow circle:** shows the click made in that step.

Simulated data and real data

ProteinDelta works in two modes. **Simulated mode** (the ⚡ Analyze WT vs Mutant button) generates the profiles with a deterministic engine from the sequence; it is useful for exploring the interface and the logic, but its values must not be cited as results. **Real mode** (the 📁 Import ColabFold section) uses the files produced by AlphaFold2/ColabFold and BioPython. The application always shows which mode is active (the “simulated” tag in the context bar).

2. General interface

When ProteinDelta opens, it shows the header with the application name, the navigation bar and, below it, the active section. The first section displayed is “Oncology Proteins”.

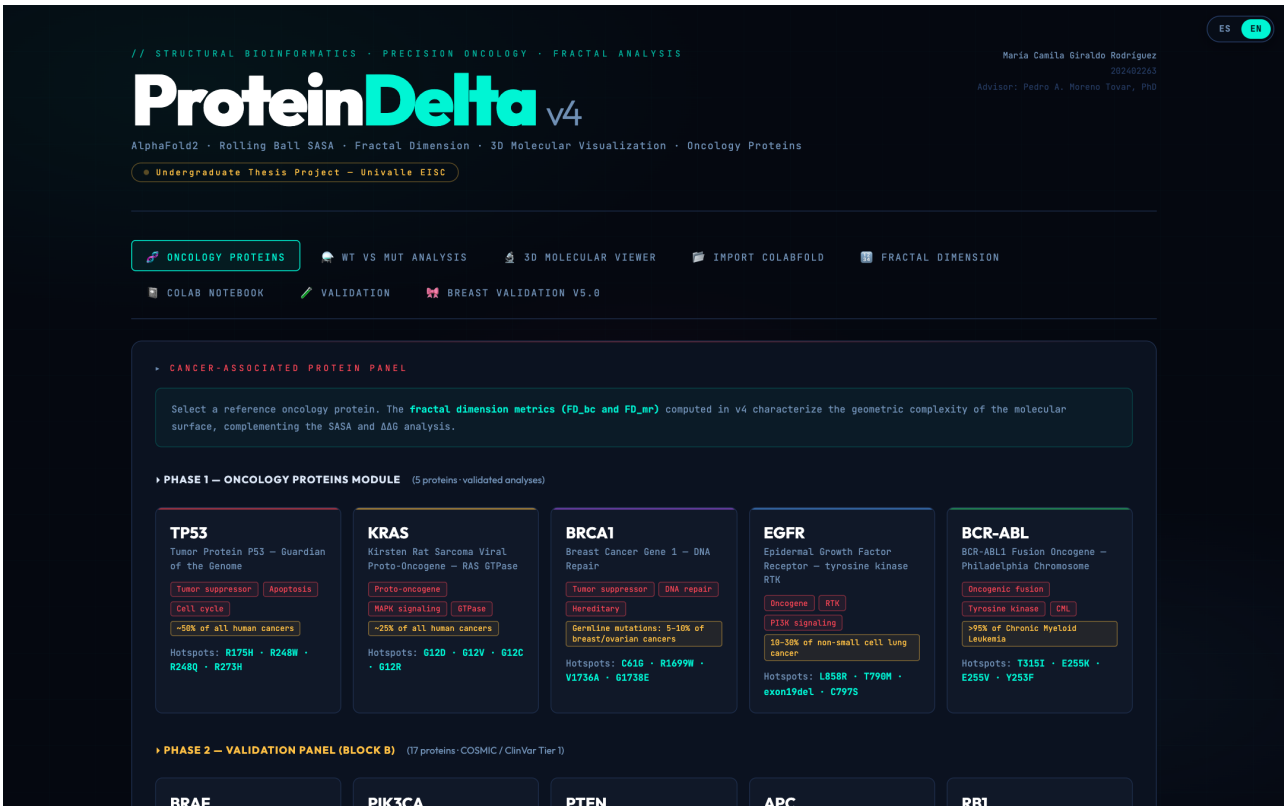


Figure 1. ProteinDelta start screen.

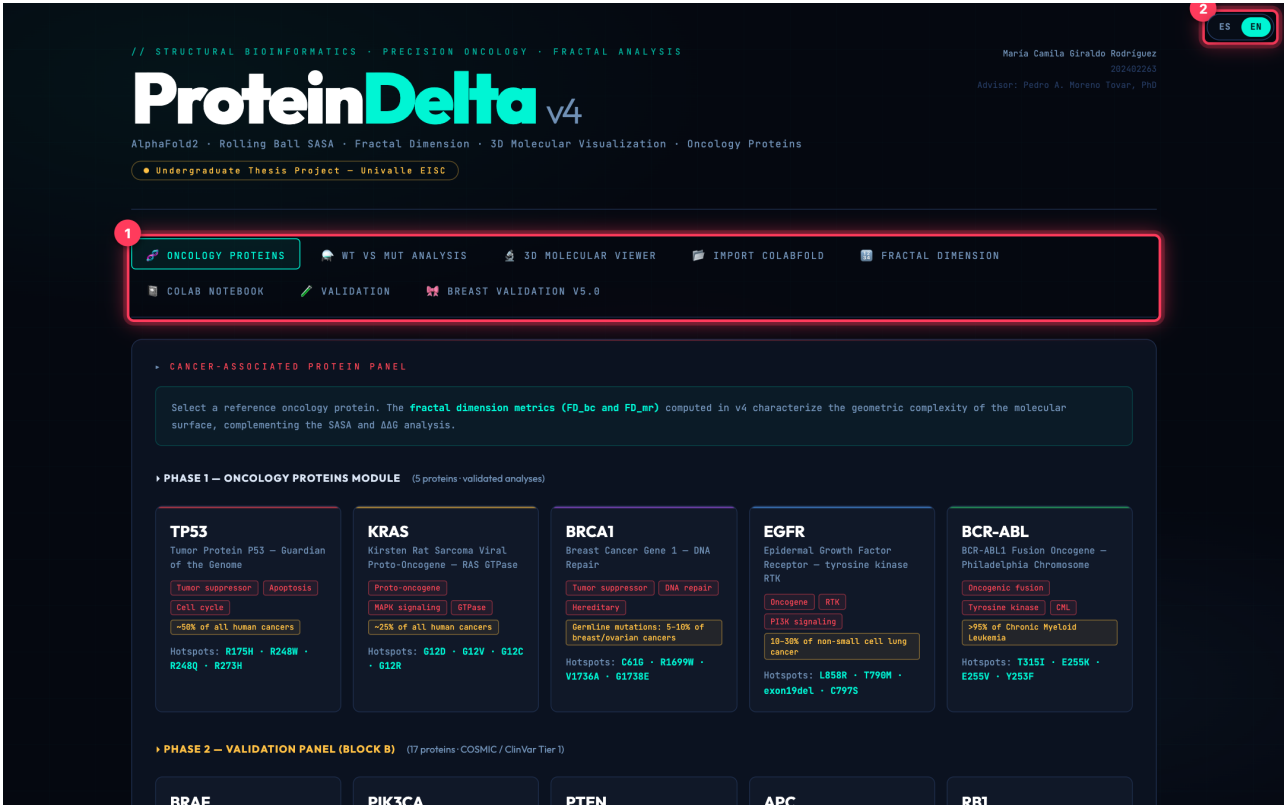


Figure 2. Navigation bar and language selector.

1 Navigation bar. Eight buttons, one per section. The button of the active section is highlighted in green. You can switch sections at any time; loaded data is kept.

2 ES / EN language selector. Switches the whole interface between Spanish and English (see chapter 11).

Section	What it is for
 Oncology Proteins	Catalog of 22 cancer-associated proteins with their canonical mutations; the starting point for an analysis.
 WT vs MUT Analysis	Sets up the sequence and mutations, runs the analysis and shows metrics, charts and a per-residue table.
 3D Molecular Viewer	Displays WT and mutant PDB structures in 3D, with styles, colors and microenvironment highlighting.
 Import ColabFold	Loads the real files generated in Google Colab and analyzes them as real data.
 Fractal Dimension	Explains and shows the global and local fractal dimension of the latest analysis.
 Colab Notebook	Downloads the ColabFold notebook, with instructions for running it.
 Validation	Pipeline validation results: replicates, benign controls, 40-pair panel.
 Breast Validation v5.0	Breast-cancer module: 40 cases + 12 controls and the spherical fractality term.

3. Oncology Proteins

This section presents a catalog of reference proteins organized in two phases: Phase 1 (5 proteins with validated analyses: TP53, KRAS, BRCA1, EGFR and BCR-ABL) and Phase 2 (17 proteins of the validation panel, Block B). Each card shows the gene, its full name, functional tags, its frequency in cancer and its hotspot mutations.

3.1 Select a protein

Step 1. Click the card of the protein you are interested in (TP53 in the example).

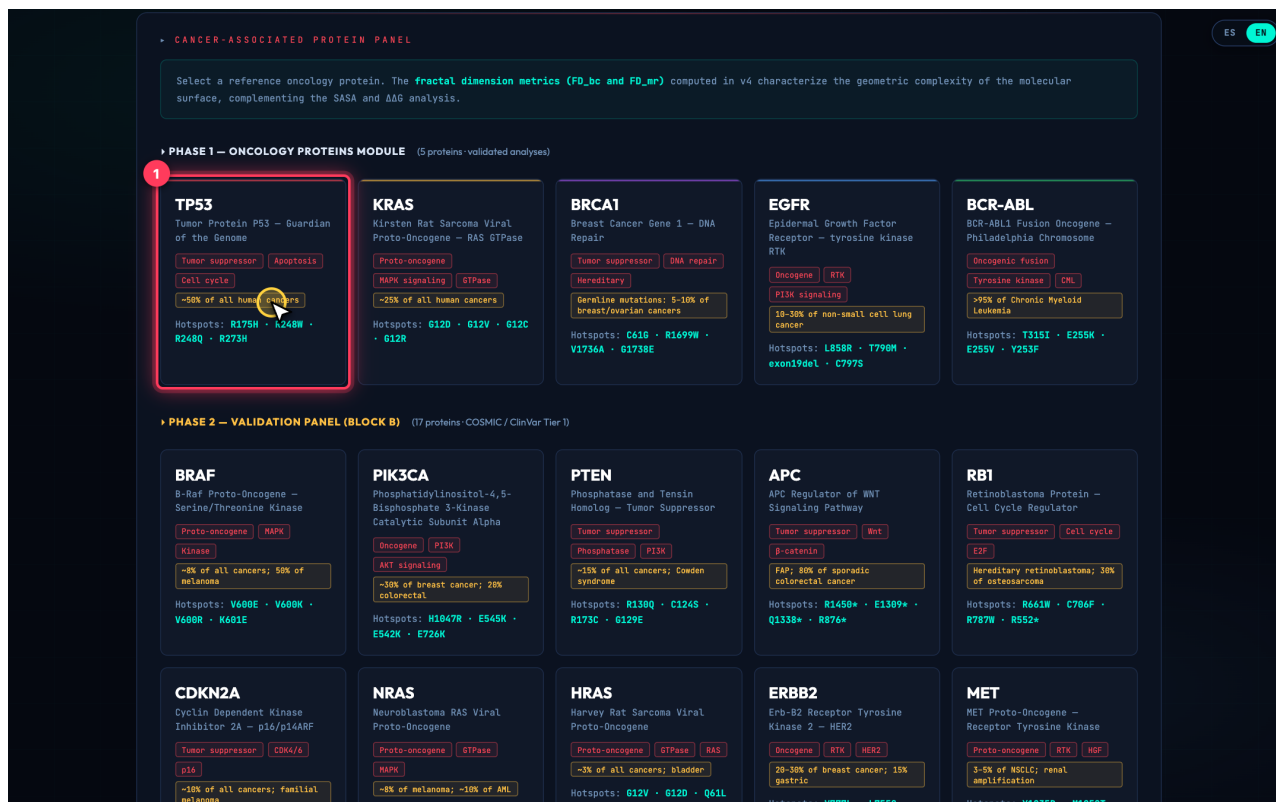


Figure 3. Clicking the TP53 card in the catalog.

1 Protein card. When clicked, it is highlighted and its detail panel opens below the catalog.

3.2 Detail panel and mutations

The detail panel shows the UniProt identifier, the PDB code, the associated cancers, the oncological context, key references and a fragment of the sequence.

Step 1. Under “Canonical mutations”, click the mutation you want to study (for example R175H). You can select several; clicking again deselects it.

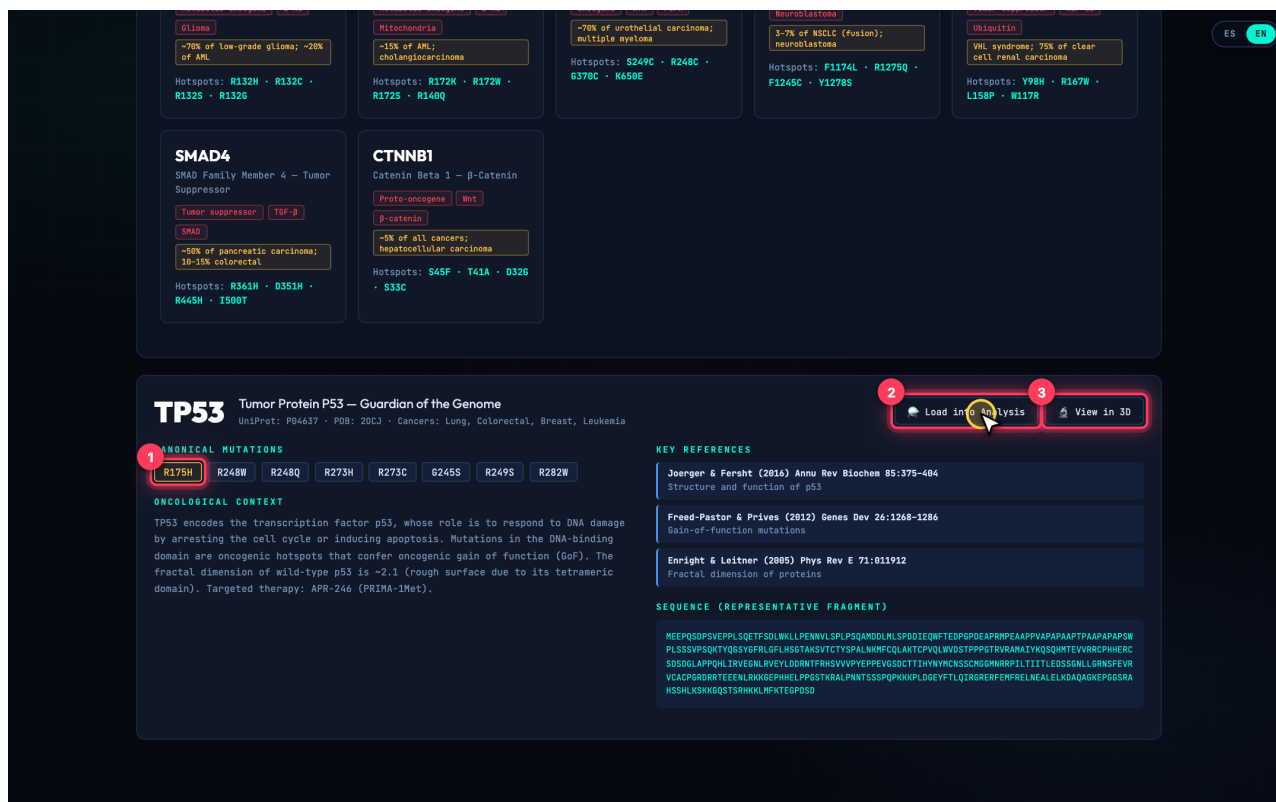


Figure 5. Selected mutation and the buttons to continue.

- 1 Selected mutation (R175H).
- 2 **Load into Analysis:** copies the sequence, the name and the mutations into the analysis form and switches section.
- 3 **View in 3D:** loads the reference PDB (2OCJ for TP53) into the 3D viewer.

4. WT vs MUT Analysis

This is the core of the application. Here you define what is compared and obtain the global metrics, the pathogenicity classification and the per-residue charts.

4.1 Set up the sequences

If you arrived from “Load into Analysis”, the form is already filled in. You can also fill it in by hand:

The screenshot shows the 'WT vs MUT ANALYSIS' setup form in ProteinDelta v4. The form is titled 'SEQUENCE SETUP' and contains several input fields and a button. Red numbered circles (1-6) highlight specific elements: 1. WT (WILD-TYPE) SEQUENCE text area with a long amino acid sequence. 2. NAME / PROTEIN text field with 'TP53'. 3. MUTATION MODE dropdown menu with 'Point mutation list' selected. 4. ADD MUTATION (E.G. D10L) text field with 'D10L' entered. 5. A pill showing 'R175H' with a close button. 6. A yellow warning bar at the bottom stating 'TP53: R175H loaded — press ⚡ Analyze to continue (simulated mode, no real PDB)'. A large blue button labeled 'Analyze WT vs Mutant' is also visible.

Figure 6. Setup form with TP53 R175H loaded.

- 1 WT (Wild-Type) sequence:** amino-acid sequence of the wild-type protein (one-letter code). It is auto-filled when a PDB is uploaded in “Import ColabFold”.
- 2 Name / Protein:** label for the analysis; it appears in the exported files.
- 3 Mutation mode:** “Point mutation list”, “Full MUT sequence” (paste the mutated sequence) or “No mutation” (MUT = WT, useful as a control).
- 4 Add mutation:** type the mutation in standard notation (original residue + position + new residue, e.g. D10L) and press “+ ADD” or Enter.
- 5 Added mutations:** each one appears as a pill; the X removes it.
- 6 Context bar:** shows what is loaded and in which mode it will be analyzed (here it warns that it will be simulated mode because there is no real PDB).

4.2 Run the analysis

Step 1. Click “⚡ Analyze WT vs Mutant”. A loading indicator appears while the profiles are computed.

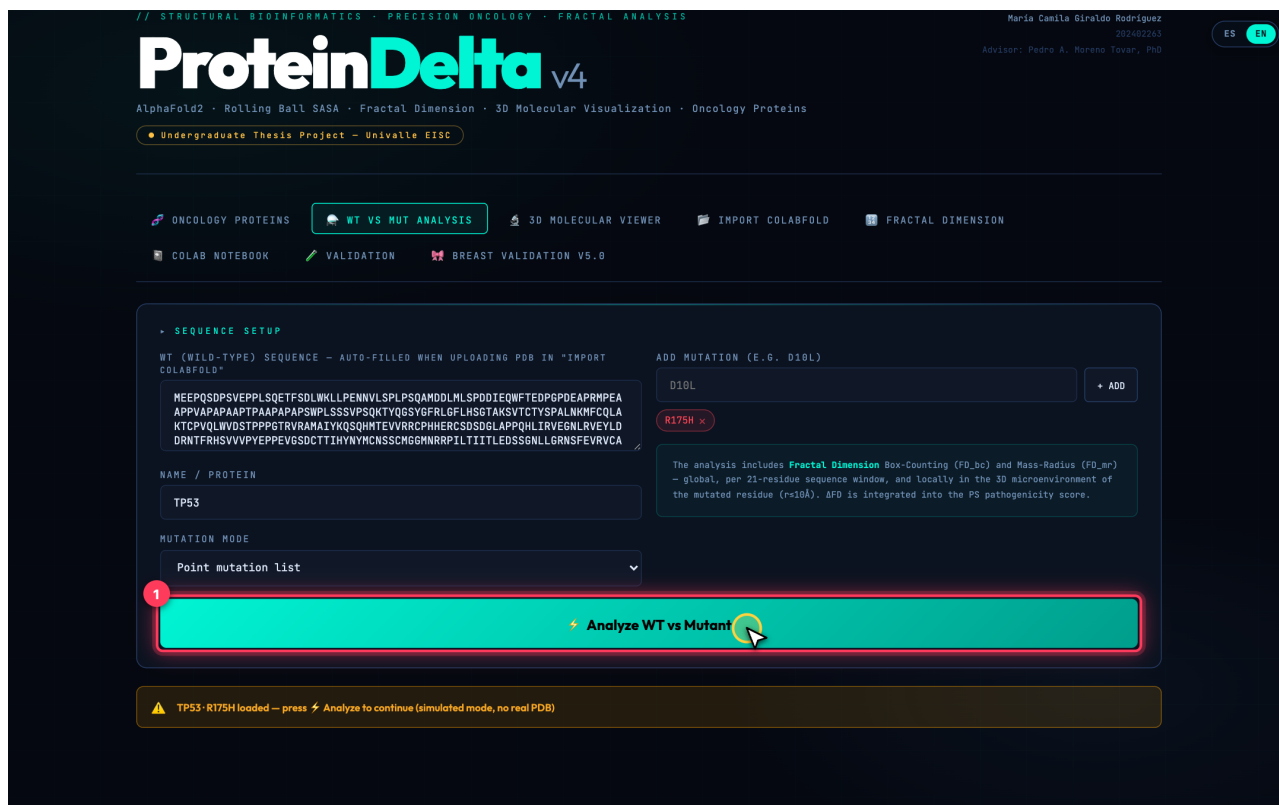


Figure 7. Button that runs the analysis.

4.3 Global metrics and classification

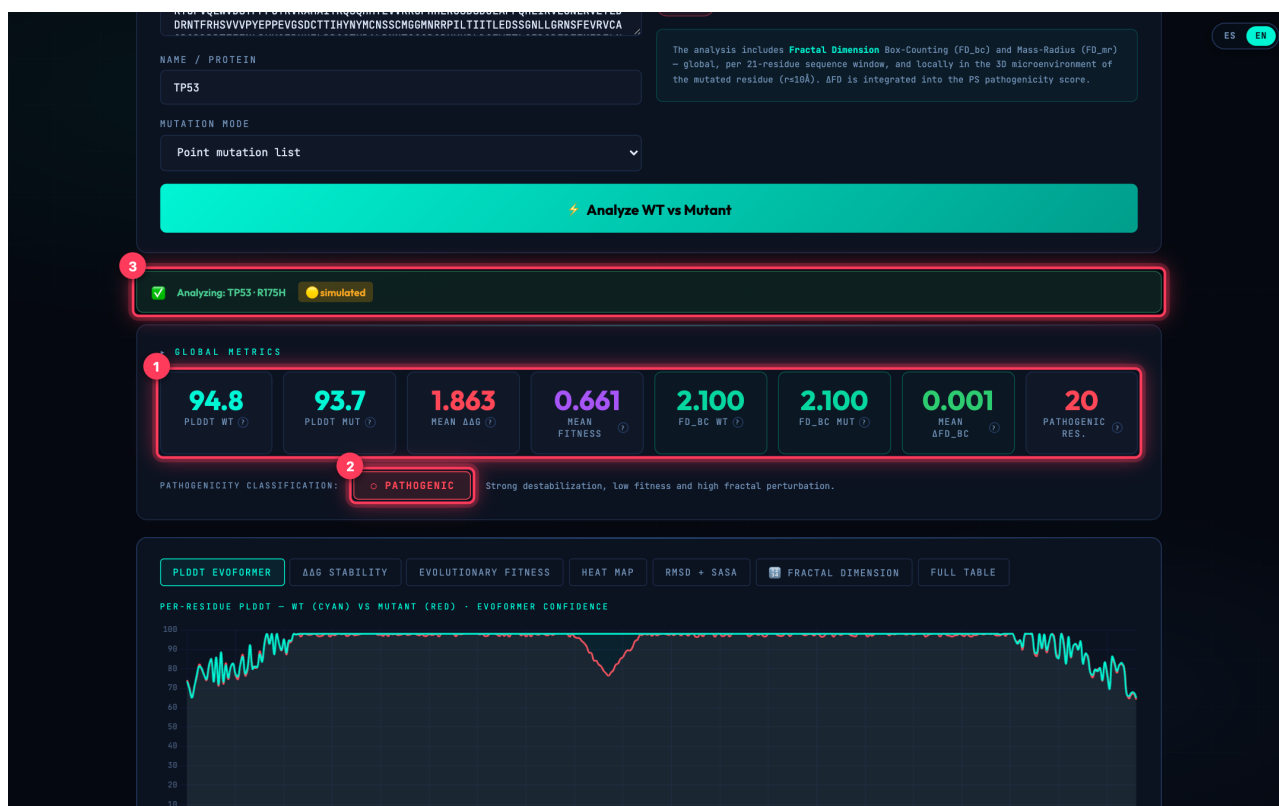


Figure 8. Result: global metrics and pathogenicity classification.

1 Global metrics: mean WT and MUT pLDDT, mean $\Delta\Delta G$, mean evolutionary fitness, WT and MUT FD_bc, mean ΔFD_bc and the number of pathogenic residues. Hover over the “?” icon of each metric to see its explanation.

- 2 Pathogenicity classification:** Pathogenic, Likely Pathogenic, Uncertain significance (VUS), Likely Benign or Benign, with a sentence summarizing why.
- 3 Context bar:** confirms the analyzed variant and the mode (here “simulated”).

4.4 Result tabs

Below the metrics there are seven tabs. Click each one to change the view:

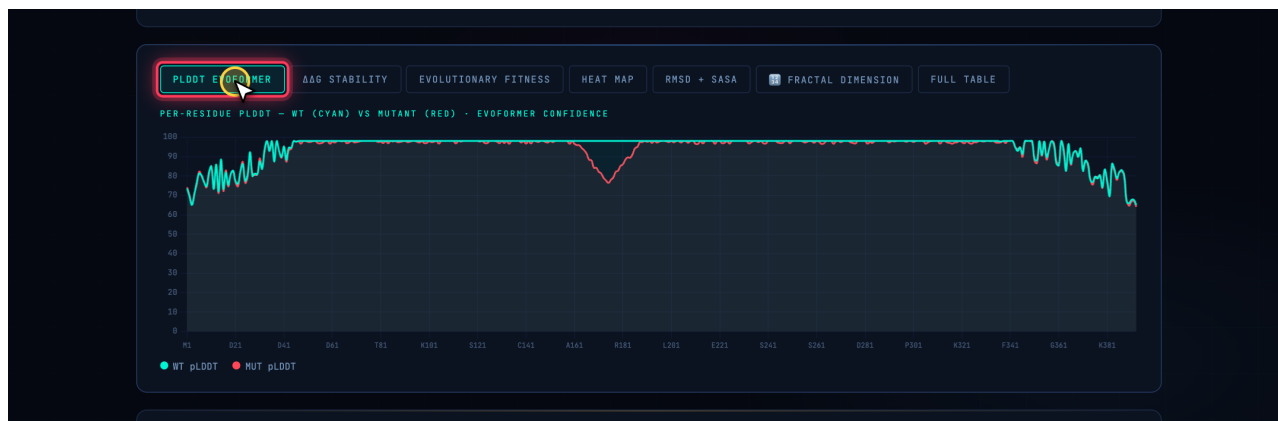


Figure 9. *pLDDT Evoformer tab: per-residue confidence, WT (cyan) vs mutant (red).*

The local drop of the red curve around the mutated position indicates a loss of structural confidence in that region.

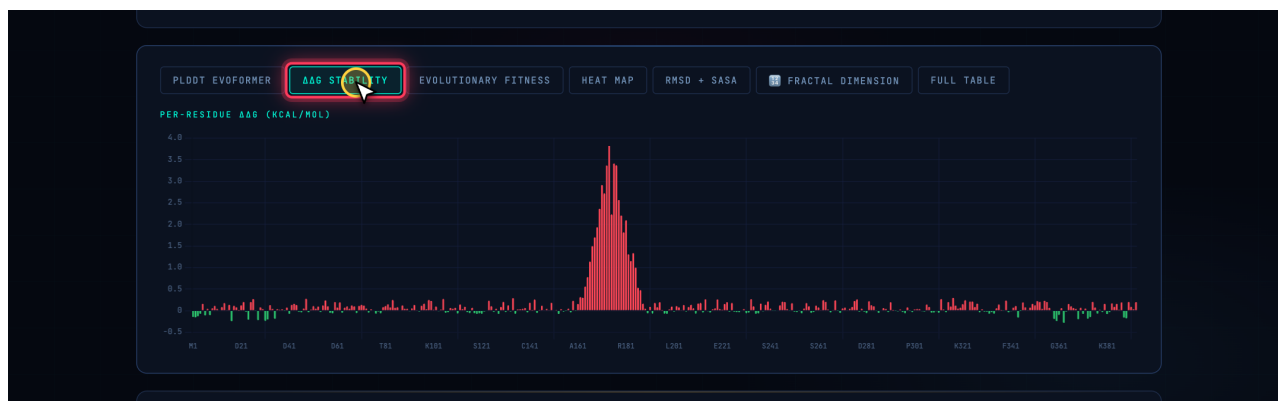


Figure 10. *ΔΔG Stability tab: per-residue change in free energy (kcal/mol); red destabilizes, green stabilizes.*



Figure 11. *Evolutionary Fitness tab: 0–1 score per position; low values (amber/red) mark positions that tolerate change poorly.*



Figure 12. Heat Map tab: estimated $\Delta\Delta G$ for all 20 possible amino acids (Y axis) at each residue (X axis).



Figure 13. RMSD + SASA tab: per-residue $C\alpha$ displacement ($\text{\AA}$) and change in accessible surface ($\text{\AA}^2$).

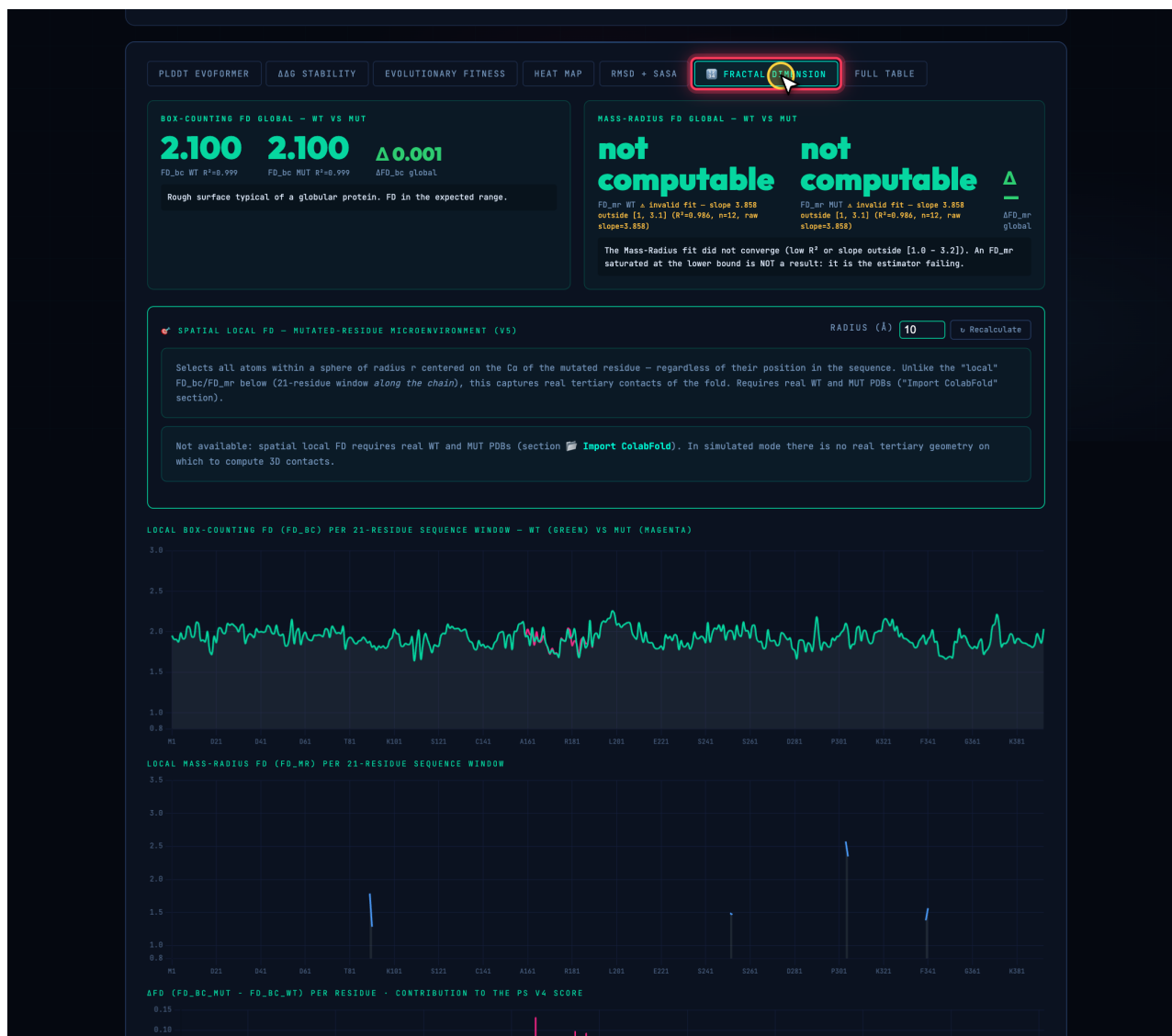


Figure 14. Fractal Dimension tab: global FD (Box-Counting and Mass-Radius), local spatial FD and the 21-residue window profile.

Local spatial FD and Mass-Radius

The local FD in the sphere around the mutated residue requires real WT and mutant PDB files (Import ColabFold section); in simulated mode it shows as “Not available”. When the Mass-Radius fit is not valid, the application shows “not computable” instead of a number: this is a safeguard, not an error.

#	AA	pLDDT WT	pLDDT MUT	ΔpLDDT	ΔAG	Fitness	FD_bc WT	FD_bc MUT	ΔFD	RMSD	ΔSASA	Pathogenicity v4
363	R	92.8	94.3	1.47	-0.286	0.869	1.7951	1.7951	0	0.0318	-2.5	Likely Benign
17	E	81.9	83	1.08	-0.248	0.813	2.0019	2.0019	0	0.0458	-3.8	Likely Benign
360	G	86.5	87.9	1.37	-0.245	0.842	1.8151	1.8151	0	0.0202	-1.3	Likely Benign
31	V	87.6	89.1	1.49	-0.244	0.849	1.9988	1.9988	0	0.0561	-0.2	Likely Benign
28	E	79.9	81.1	1.17	-0.225	0.802	2.0971	2.0971	0	0.0691	-1.9	Likely Benign
24	K	86.2	87.3	1.18	-0.214	0.843	1.9843	1.9843	0	0.0634	-1.3	Likely Benign
32	L	83.6	84.6	0.94	-0.209	0.827	2.073	2.073	0	0.0515	-2.9	Likely Benign
369	L	89.8	90.9	1.05	-0.203	0.863	1.8684	1.8684	0	0.0311	-1.8	Likely Benign
359	P	96	97	0.93	-0.197	0.89	1.8575	1.8575	0	0.0677	-2.4	Likely Benign
35	L	92.8	93.9	1.15	-0.193	0.878	1.9342	1.9342	0	0.0459	-0.4	Likely Benign
373	K	86	86.7	0.72	-0.189	0.842	2.0785	2.0785	0	0.0682	-3.7	Likely Benign
389	G	67	67.7	0.7	-0.184	0.691	1.8428	1.8428	0	0.0224	-3.6	Likely Benign
388	E	79.8	80.9	1.13	-0.166	0.807	1.8069	1.8069	0	0.0419	0.8	Likely Benign
344	L	92.9	93.9	1.09	-0.165	0.881	1.8728	1.8728	0	0.0683	0.5	Likely Benign
2	E	69.5	70.2	0.75	-0.16	0.719	1.9062	1.9062	0	0.0605	-2	Likely Benign
1	M	73.6	74.2	0.56	-0.154	0.756	1.9526	1.9526	0	0.0362	-3.2	Likely Benign
374	G	78.1	78.6	0.54	-0.153	0.791	1.8134	1.8134	0	0.0296	-3.3	Likely Benign
3	E	65.1	65.5	0.34	-0.133	0.673	1.9036	1.9036	0	0.0546	-4	Likely Benign
6	S	81.3	82.3	1	-0.115	0.822	1.9115	1.9115	0	0.0311	2.2	Likely Benign
8	P	76.8	77.1	0.25	-0.11	0.784	2.0655	2.0655	0	0.0565	-3.5	Likely Benign
361	G	85.4	85.8	0.46	-0.104	0.846	1.793	1.793	0	0.0361	-1.5	Likely Benign
381	K	76.8	77.9	1.03	-0.085	0.793	1.9114	1.9114	0	0.0467	4	Likely Benign
371	S	82.6	82.7	0.13	-0.081	0.828	2.0956	2.0956	0	0.0504	-3	Likely Benign
153	P	98	98	0	-0.08	0.908	1.9609	1.9609	0	0.063	-4	Likely Benign
119	A	98	98	0	-0.079	0.908	1.8357	1.8357	0	0.048	-3.9	Likely Benign
128	P	98	98	0	-0.078	0.908	2.0373	2.0373	0	0.0639	-3.9	Likely Benign
198	E	98	98	0	-0.078	0.908	2.1697	2.1697	0	0.0418	-3.9	Likely Benign
4	P	70.6	71	0.44	-0.077	0.734	1.8749	1.8749	0	0.0269	-0.3	Uncertain significance
41	D	95.2	95.3	0.15	-0.077	0.897	2.0014	2.0014	0	0.0405	-2.6	Likely Benign
121	S	98	98	0	-0.076	0.908	1.8229	1.8229	0	0.0663	-3.8	Likely Benign
135	C	98	98	0	-0.076	0.908	1.8805	1.8805	0	0.0237	-3.8	Likely Benign
332	I	98	98	0	-0.076	0.908	1.8944	1.8944	0	0.0585	-3.8	Likely Benign
235	N	98	98	0	-0.075	0.908	1.8483	1.8483	0	0.0697	-3.7	Likely Benign
77	P	98	98	0	-0.073	0.908	1.9771	1.9771	0	0.0438	-3.7	Likely Benign
197	V	98	98	0	-0.073	0.908	2.1589	2.1589	0	0.0695	-3.7	Likely Benign
377	T	78.7	79.6	0.83	-0.073	0.807	1.8934	1.8934	0	0.03	3	Uncertain significance

Figure 15. Full table tab: every metric per residue, sortable.

In the table, **click a column header** (those marked ↕) to sort by it; clicking again reverses the order. The “?” icon on each header explains the metric.

4.5 Export the results

165	Q	98	93.2	-4.81	0.779	0.809	1.9348	1.9981	0.0633	0.5159	0.5	Uncertain significance
185	S	98	91.3	-6.73	1.001	0.776	1.804	1.8976	0.0935	0.5535	-3.8	Likely Pathogenic
166	S	98	91.5	-6.52	1.139	0.762	1.8782	1.8761	-0.002	0.4298	4.8	Likely Pathogenic
183	S	98	89.6	-8.36	1.15	0.753	1.8454	1.8516	0.0063	0.7385	-9.4	Likely Pathogenic
182	C	98	88.4	-9.61	1.306	0.73	2.0367	1.8871	-0.1496	0.9832	-11.6	Likely Pathogenic
184	D	98	89.4	-8.58	1.335	0.731	1.6811	1.6861	0.005	0.2736	-1.9	Likely Pathogenic
167	Q	98	88.4	-9.56	1.498	0.709	1.9158	1.877	-0.0388	0.3299	-1.6	Likely Pathogenic
168	H	98	88.1	-9.87	1.701	0.686	2.0562	1.9477	-0.1085	0.8623	6.1	Likely Pathogenic
180	E	98	85.2	-12.79	1.813	0.66	1.9214	2.0195	0.0981	0.4197	-11.7	Pathogenic
169	M	98	84.2	-13.8	1.936	0.642	1.8719	1.8719	0	0.6698	-13.6	Pathogenic
181	R	98	86	-12.04	2.096	0.633	1.9381	1.829	-0.1091	0.4795	8.5	Pathogenic
179	H	98	82.8	-15.22	2.205	0.607	2.0294	2.0396	0.0102	0.9384	-11.5	Pathogenic
175	R	98	76.4	-21.64	2.23	0.569	1.6847	1.6881	0.0035	0.9086	-61.6	Pathogenic
170	T	98	84.2	-13.78	2.364	0.597	1.8095	1.7992	-0.0103	1.2491	8	Pathogenic
178	H	98	82.1	-15.85	2.564	0.566	1.8862	1.8881	0.002	1.7619	1.4	Pathogenic
172	V	98	81.3	-16.75	2.718	0.546	1.7746	1.7934	0.0188	1.1331	1.9	Pathogenic
171	E	98	82.2	-15.77	2.911	0.53	1.7343	1.7244	-0.0099	0.3338	19.4	Pathogenic
173	V	98	79	-19.01	3.364	0.469	1.7285	1.7355	0.007	0.6713	16.1	Pathogenic
177	P	98	79.2	-18.8	3.37	0.47	1.8902	1.897	0.0068	0.9214	18	Pathogenic
176	C	98	78	-19.97	3.409	0.46	1.9075	1.9178	0.0104	0.9911	10.7	Pathogenic
174	R	98	77.7	-20.32	3.819	0.418	1.7318	1.7387	0.0069	1.8623	28.4	Pathogenic

EXPORT V4 RESULTS

1

↓ full_analysis.csv

2

↓ results.json

3

📦 all_outputs.zip

Figure 16. Export buttons at the end of the section.

- 1 full_analysis.csv:** per-residue table, ready for Excel, R or Python.
- 2 results.json:** complete results in a structured format.
- 3 all_outputs.zip:** a bundle with every file of the analysis.

5. 3D Molecular Viewer

The viewer shows the WT structure (left) and the mutant structure (right) side by side. Drag with the mouse to rotate, use the wheel to zoom and right-click-drag to pan.

Step 1. Click “🔬 3D Molecular Viewer” in the navigation bar.

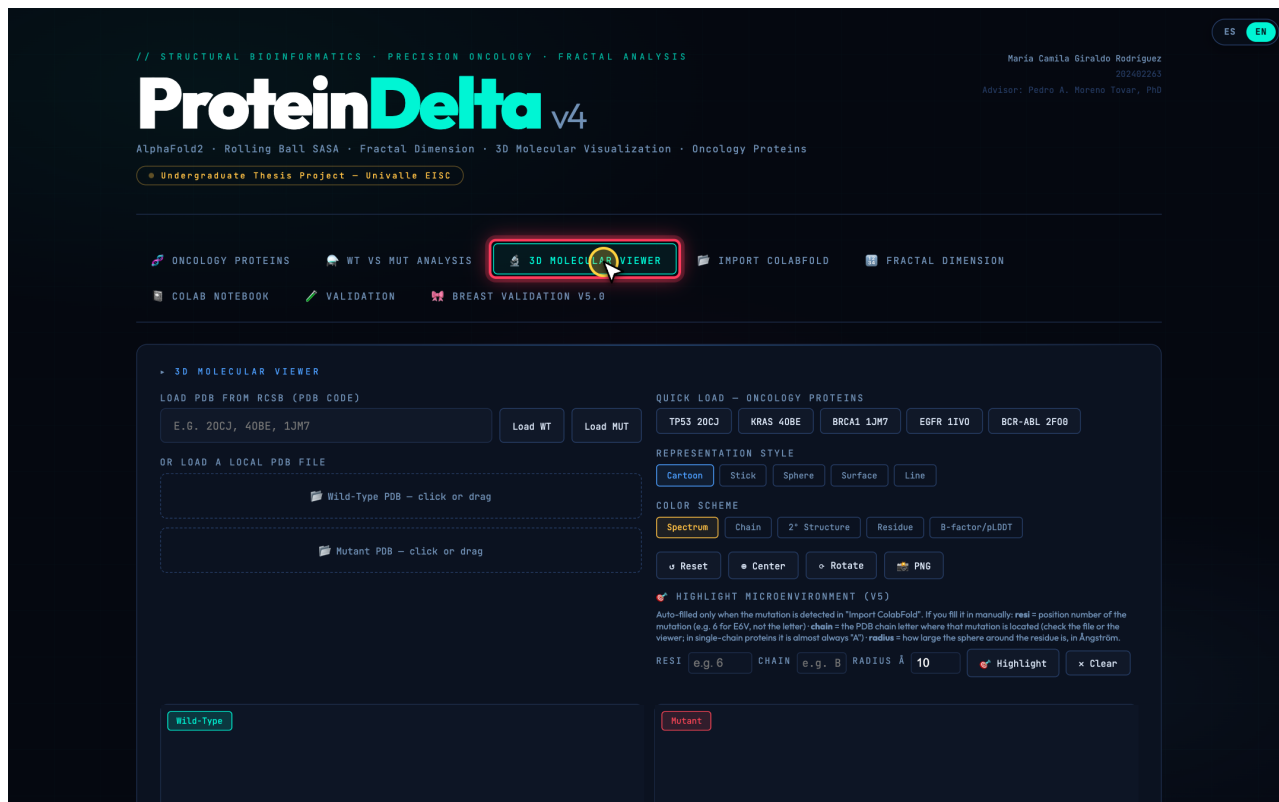


Figure 17. Opening the 3D viewer from the navigation bar.

5.1 Load a structure

There are three ways to load a structure:

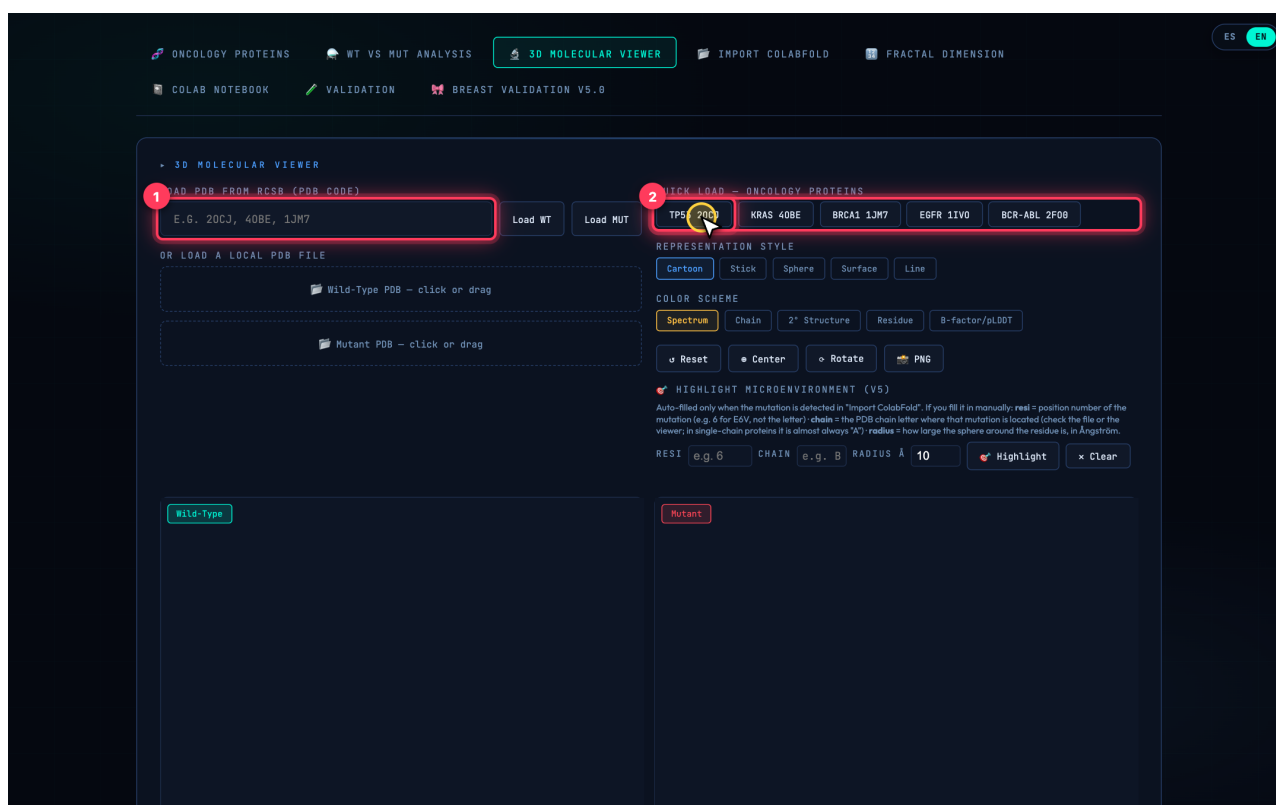


Figure 18. Viewer loading options.

- 1 **PDB code:** type a 4-character code (e.g. 2OCJ) and click “Load WT” or “Load MUT” to download it from RCSB into the corresponding panel.
- 2 **Quick load:** buttons with the Phase 1 reference structures (TP53, KRAS, BRCA1, EGFR, BCR-ABL); they load into the WT panel.

The third option is the “Wild-Type PDB” and “Mutant PDB — click or drag” areas, which accept a local .pdb file (for example, the ColabFold rank_001 models).

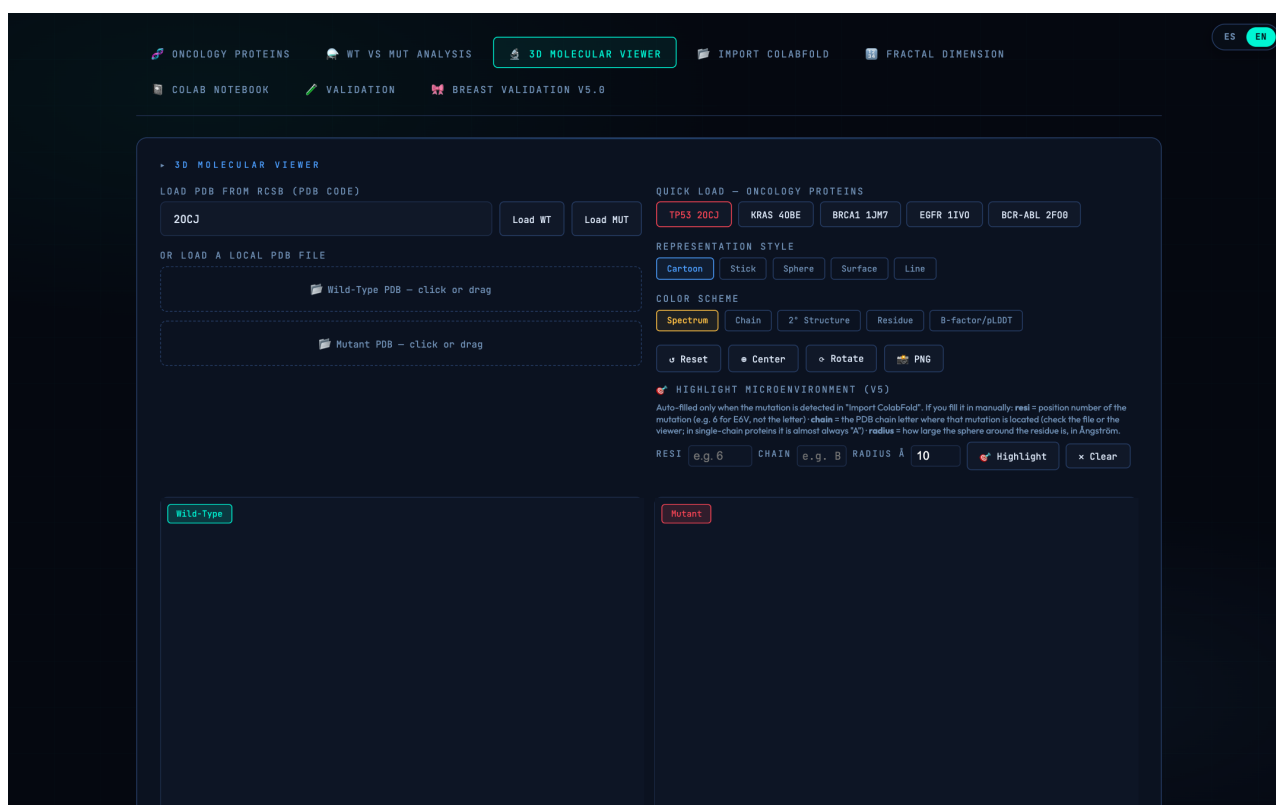


Figure 19. Structure 20CJ (TP53) loaded in the WT panel with the Cartoon style and Spectrum colors.

5.2 Style and color

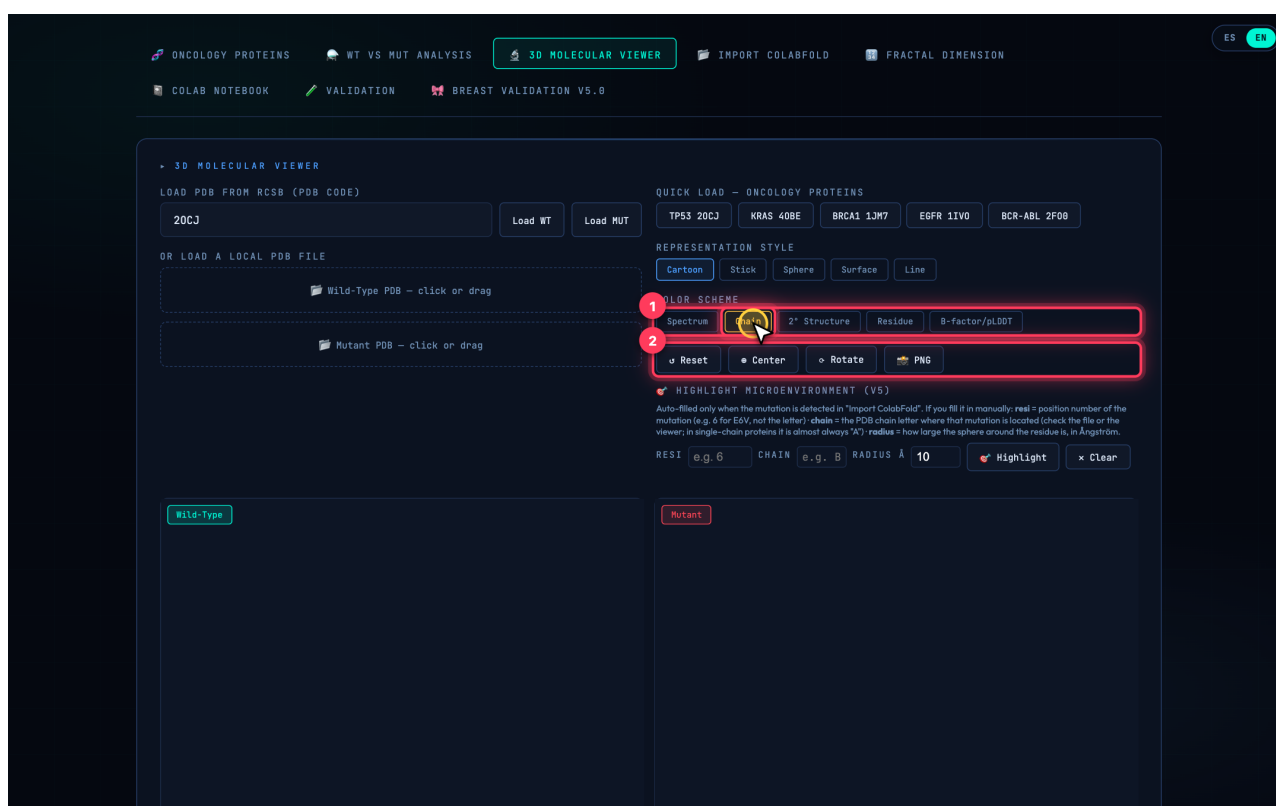


Figure 20. Changing the color scheme to “Chain”.

- 1 Color scheme:** Spectrum (rainbow from N- to C-terminus), Chain, 2° Structure (helices/sheets), Residue (by amino acid) and B-factor/pLDDT (in AlphaFold models, red = low confidence, blue = high).

2 View controls: “Reset” and “Center” reframe the molecule, “Rotate” turns on automatic spinning and “PNG” downloads an image of the view.

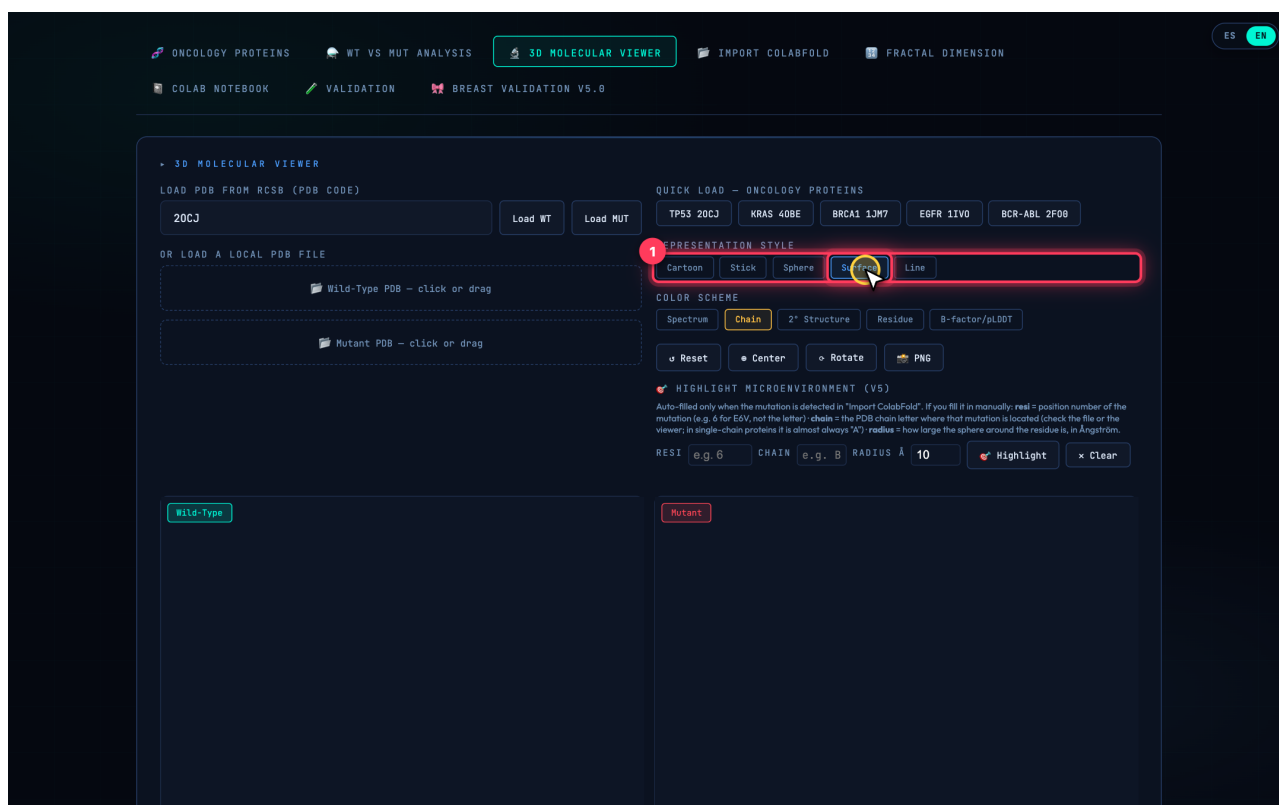


Figure 21. Changing the representation style to “Surface”.

1 Representation style: Cartoon (secondary-structure backbone), Stick (bonds), Sphere (atoms), Surface (molecular surface) and Line (lines).

5.3 Highlight the mutation microenvironment

This tool marks the mutated residue and every residue within a given radius, so you can see which 3D contacts the mutation might disturb.

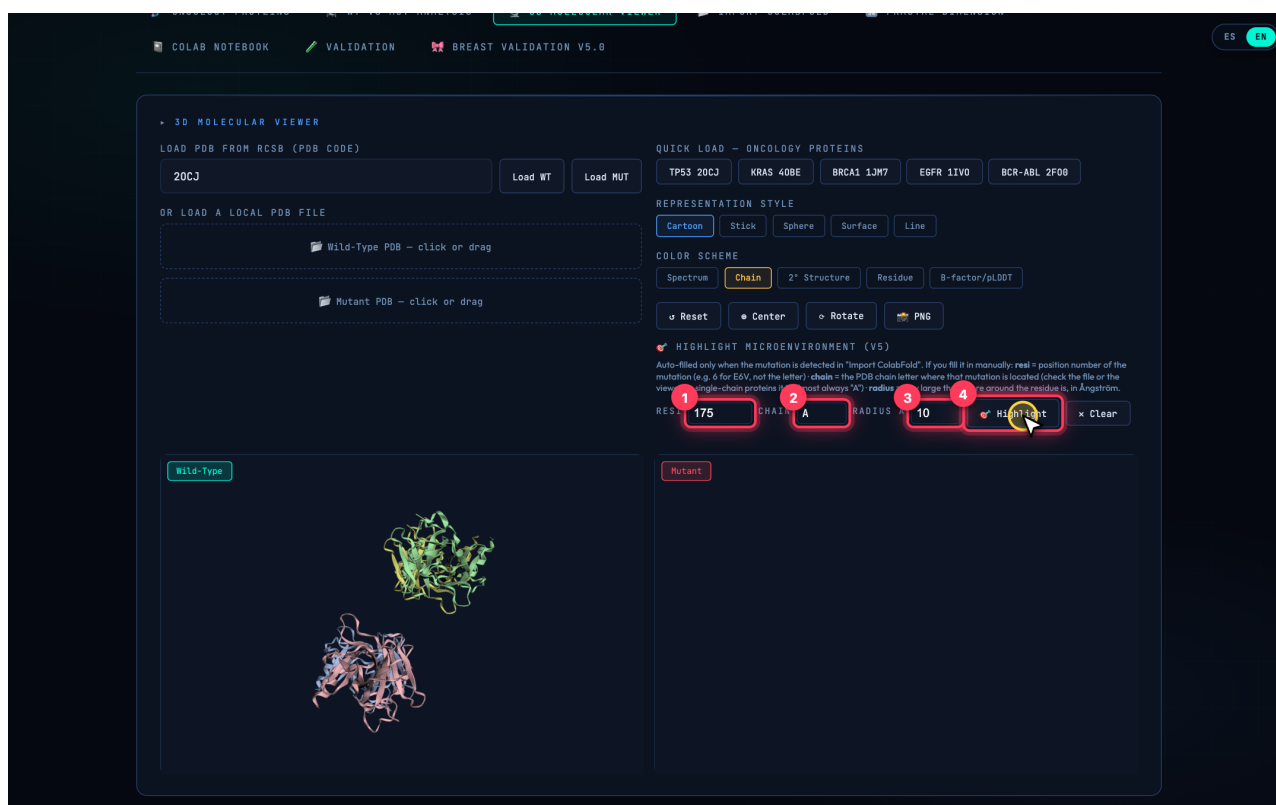


Figure 22. Highlighting residue 175 of chain A with a 10 Å radius.

- 1 **resi:** position number of the mutation (175 for R175H, without the letters).
- 2 **chain:** letter of the PDB chain that carries the mutation (almost always “A” for single-chain proteins).
- 3 **radius Å:** size of the sphere around the residue, in ångström.
- 4 **Highlight:** applies the highlight; “× Clear” removes it.

When files are imported in “Import ColabFold”, these fields are filled in automatically from the detected mutation.

6. Import ColabFold (real data)

This section turns ProteinDelta into a real-data analysis tool. It takes the files generated by the Google Colab notebook (chapter 8), which come inside `all_outputs.zip`.

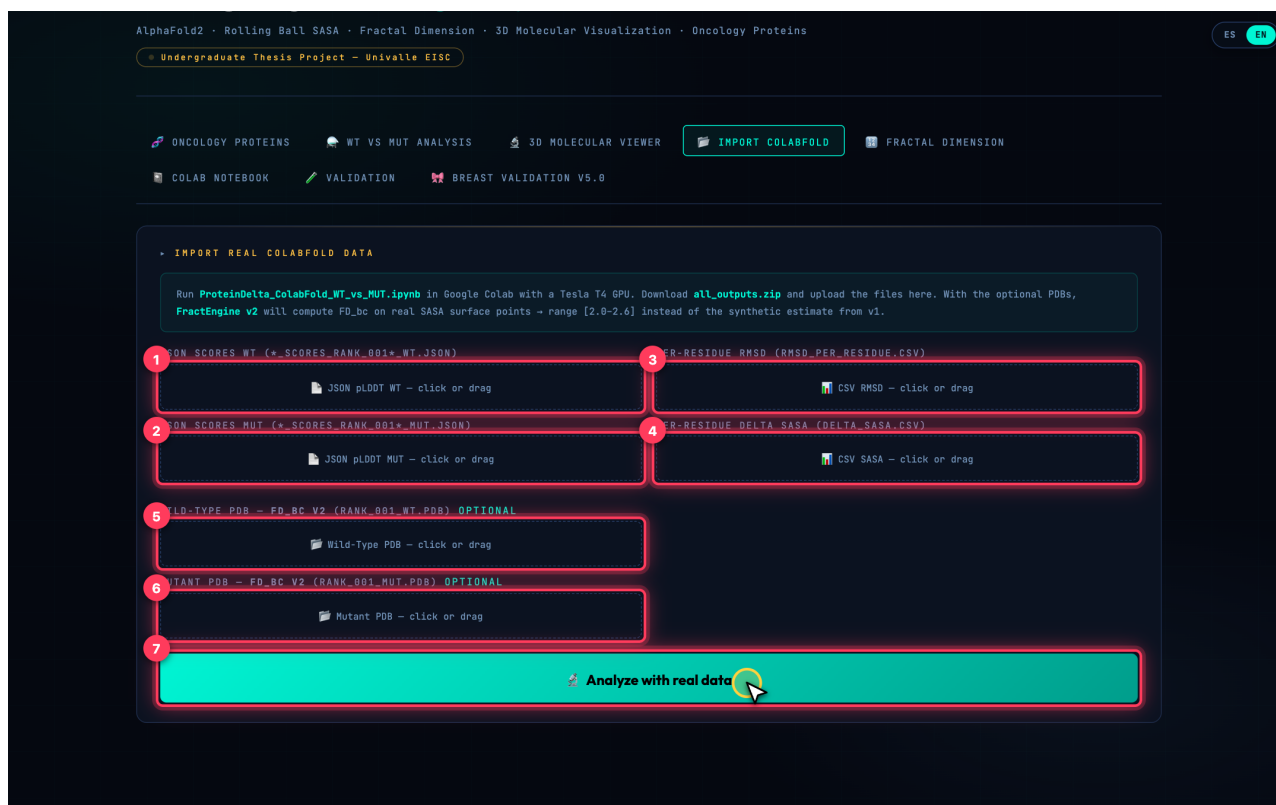


Figure 23. File upload areas and the real-data analysis button.

- 1 WT scores JSON** (*_scores_rank_001*_WT.json): per-residue pLDDT of the wild-type model.
- 2 MUT scores JSON** (*_scores_rank_001*_MUT.json): per-residue pLDDT of the mutant model.
- 3 Per-residue RMSD** (rmsd_per_residue.csv): C α displacement after Kabsch superposition.
- 4 Delta SASA** (delta_sasa.csv): per-residue change in accessible surface (Shrake-Rupley).
- 5 Wild-Type PDB** (rank_001_WT.pdb, optional): enables FD_bc v2 on the real surface and the local spatial FD.
- 6 Mutant PDB** (rank_001_MUT.pdb, optional): the same for the mutant.
- 7 Analyze with real data**: runs the analysis and shows the results in “WT vs MUT Analysis”.

6.1 Step by step

- Step 1.** Unzip `all_outputs.zip` on your computer.
- Step 2.** For each area, click it and choose the file, or drag the file onto it from your file browser. The area turns green once it receives the file.
- Step 3.** Click “🖱️ Analyze with real data”.
- Step 4.** Review the results in the analysis section (chapter 4); the context bar will indicate that they are real data. The PDB files also become available in the 3D viewer.

Required files

rmsd_per_residue.csv and **delta_sasa.csv** are mandatory for a real-data analysis. If either is missing, the application shows a warning and does not run the analysis: it does not compute RMSD or Δ SASA on its own and does not fill in missing values. If no file is loaded at all, the button runs the analysis in simulated mode.

7. Fractal Dimension

The fractal dimension (FD) quantifies the geometric complexity of a protein's surface and backbone. ProteinDelta computes two complementary estimators: Box-Counting (FD_bc), which captures surface roughness, and Mass-Radius (FD_mr), which captures fold compactness.

The section shows the values of the latest analysis. If you have not analyzed anything yet, the cards read “Run an analysis to see the fractal dimension”.

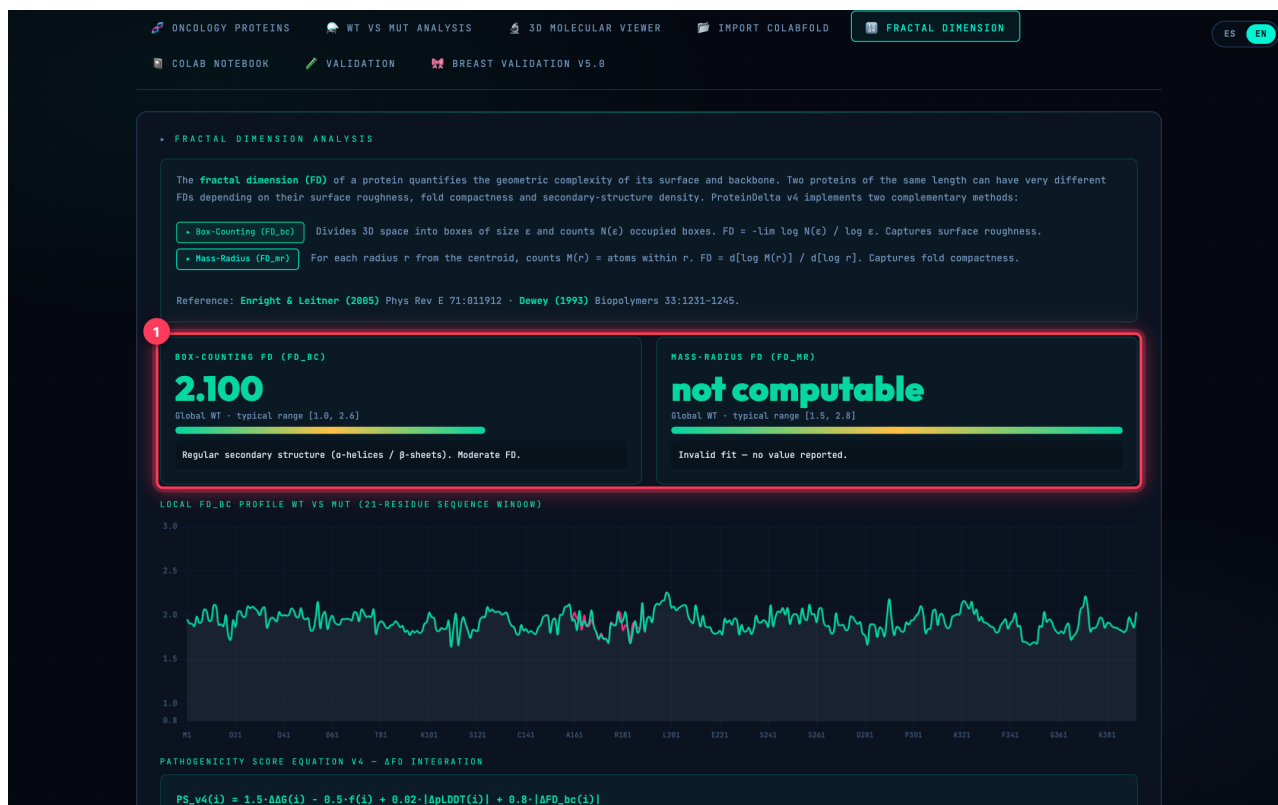
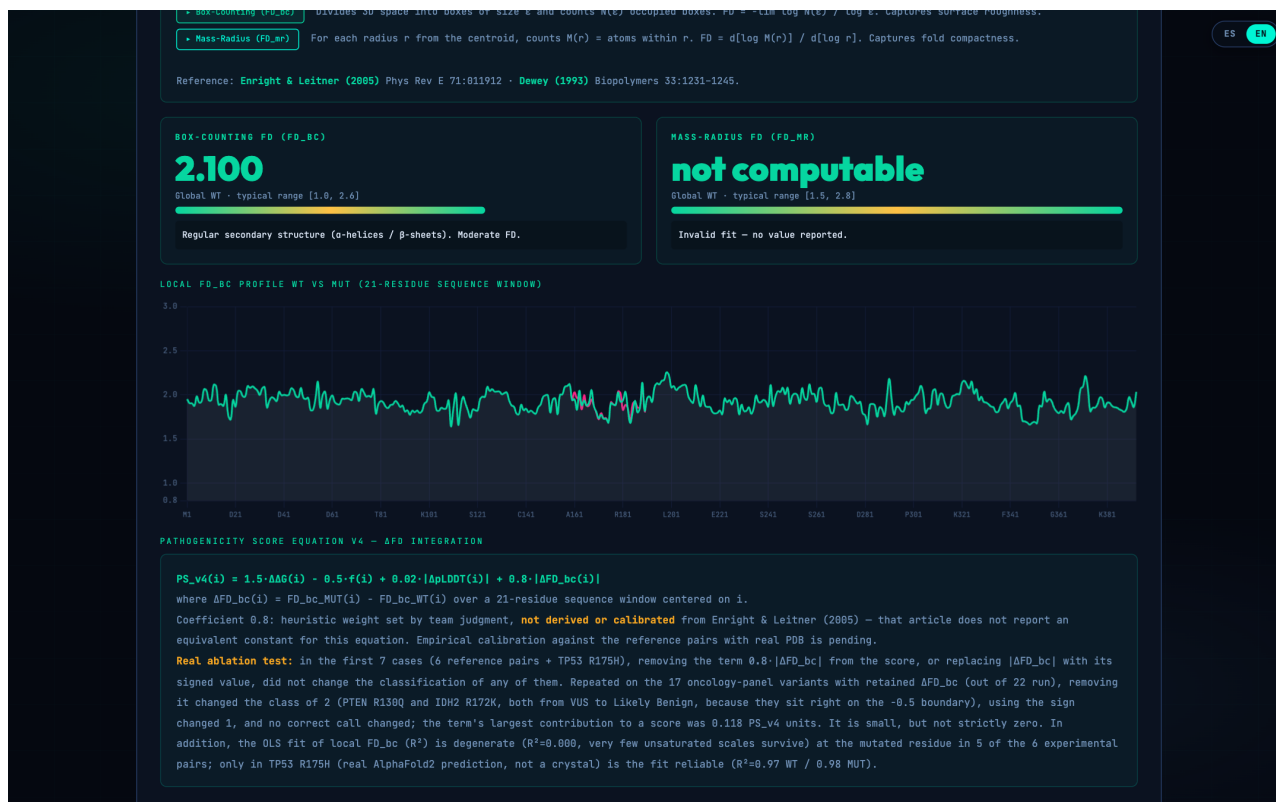


Figure 24. Global FD cards after analyzing TP53 R175H.

- 1 Global FD:** FD_bc and FD_mr of the WT protein, with their typical range and a short interpretation. “not computable” means the fit was not valid, so no number is reported.



The profile compares the local FD of WT (green) and mutant (magenta) along the sequence. Below it are the PS_v4 pathogenicity score equation and a methodological note on the weight of the fractal term and the ablation test results.

8. Colab Notebook

This is where you download the notebook that builds the WT and mutant 3D models with ColabFold (AlphaFold2) and computes per-residue RMSD and SASA. Its output is the input of “Import ColabFold”.

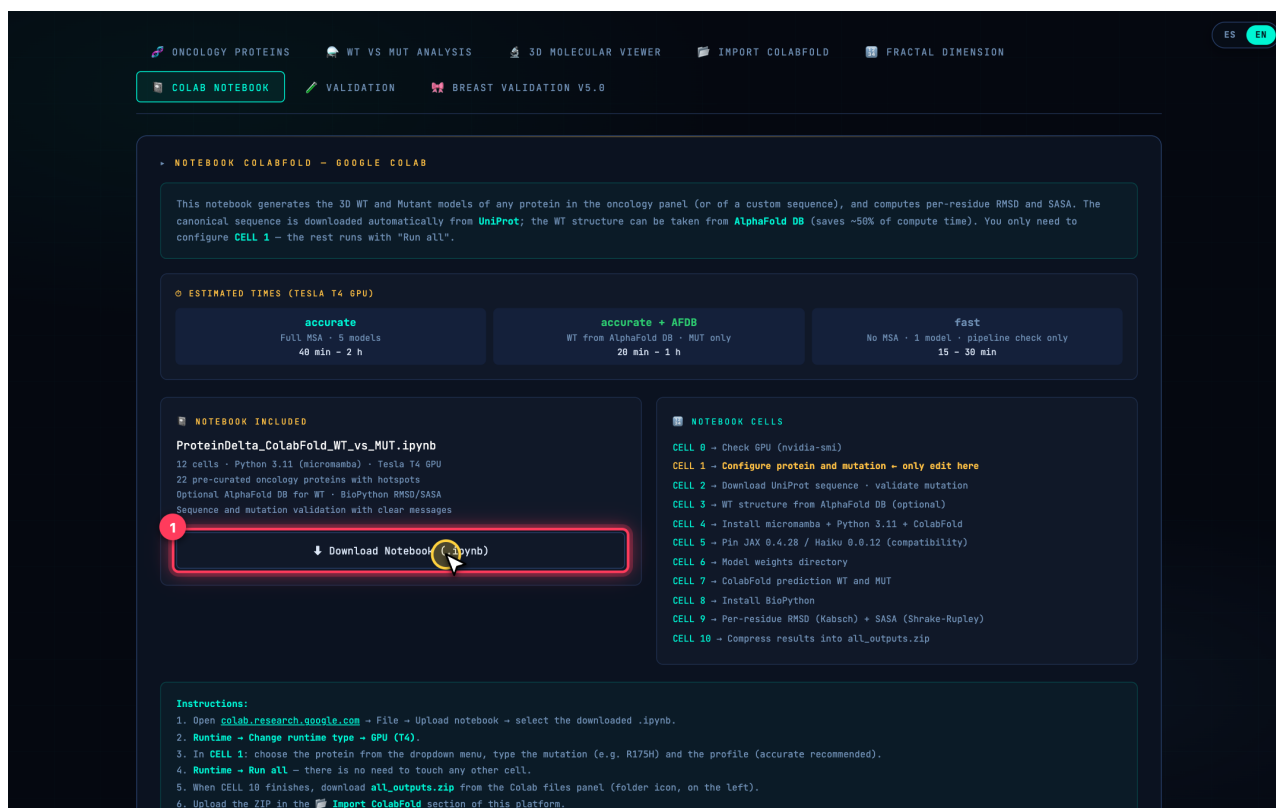


Figure 26. Notebook section: estimated times, download and cells.

- 1 **Download Notebook (.ipynb):** downloads ProteinDelta_ColabFold_WT_vs_MUT.ipynb.

8.1 How to run it

- Step 1.** Open colab.research.google.com → File → Upload notebook → select the downloaded .ipynb.
- Step 2.** Runtime → Change runtime type → GPU (T4).
- Step 3.** In CELL 1, pick the protein from the menu, type the mutation (e.g. R175H) and choose the profile. It is the only cell you need to edit.
- Step 4.** Runtime → Run all.
- Step 5.** When CELL 10 finishes, download [all_outputs.zip](#) from the Colab file panel (folder icon).
- Step 6.** Load the files in the “Import ColabFold” section (chapter 6).

Profile	What it does	Time (T4 GPU)
accurate	Full MSA, 5 models. Recommended for results.	40 min – 2 h
accurate + AFDB	Takes the WT from AlphaFold DB; only the mutant is predicted.	20 min – 1 h
fast	No MSA, 1 model. Only for checking that the pipeline runs.	15 – 30 min

The fast profile

It skips the multiple sequence alignment, AlphaFold2's main source of evolutionary information. Its models have low pLDDT and must not be used to report results.

9. Validation

This section gathers the pipeline's validation evidence. It is mostly for reading: tables and summary cards, with “What this shows” notes that interpret each block. From top to bottom it contains:

- The score as a distribution: replicates with 3–4 seeds on 8 variants.
- Real benign controls: 11 benign variants.
- Real pipeline results: 22 variants of the oncology panel, and their replicate with a second seed.
- 40-pair panel with real data (39/40) and a replicate of 8 pairs.
- Validation panel with simulated data, validation criteria and the full panel of 40 WT/mutant pairs.

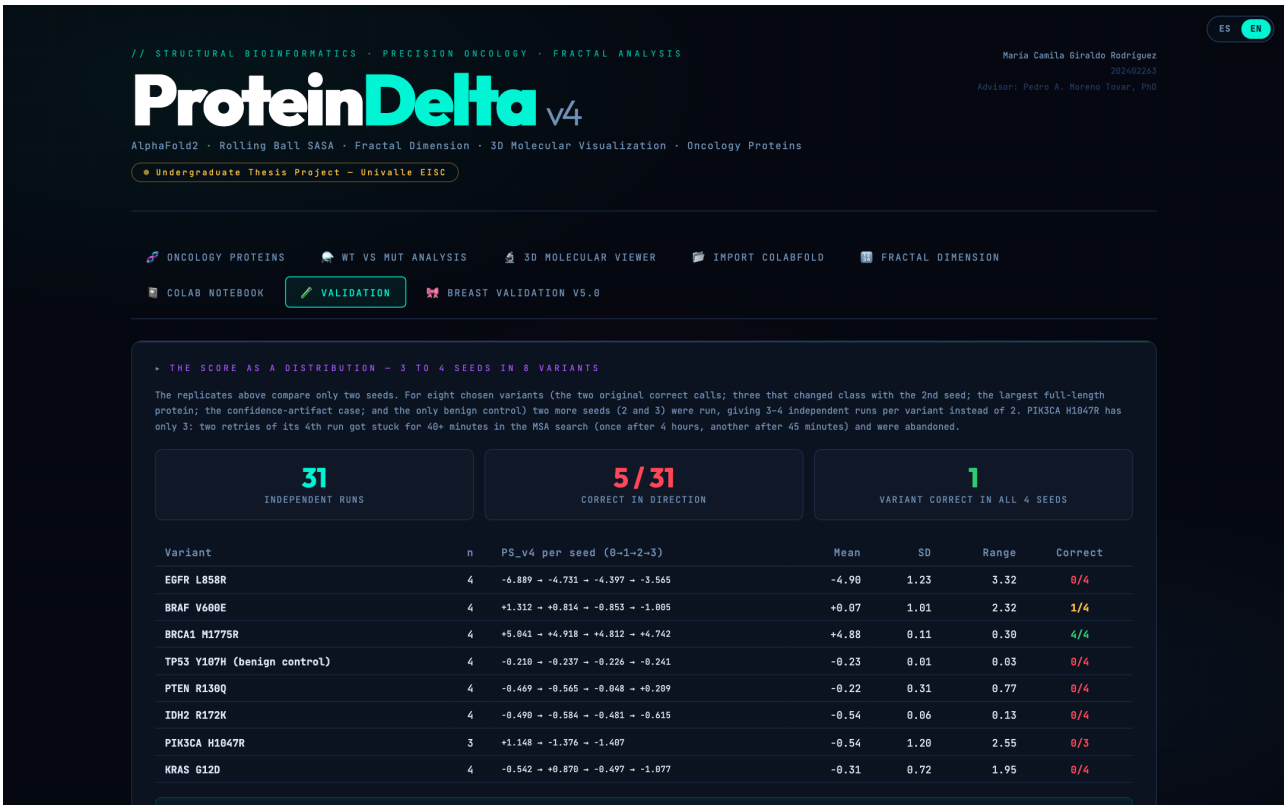


Figure 27. Top of the Validation section.

9.1 Filter the full panel and open a pair

Filters:

- AF2** Detectable structural impact Co RMSD > 0.01 Å in the mutation region RMSD > 0.01 Å in ≥ 35/40 pairs ✓ PASS - 40/40
- AF3** ASASA consistent with exposure Detectable change in surface accessibility ASASA ≠ 0 in ≥ 35/40 pairs ✓ PASS - 40/40
- FD1** Valid FD_mr range [1.0 - 3.0] FD_mr within the theoretical range for protein chains FD_mr ∈ [1.0, 3.0] in 40/40 pairs ✓ PASS - 40/40
- FD2** Inter-protein variation in FD_mr FD_mr reflects differences in structural complexity between genes Different FD_mr range in ≥ 3 protein classes ✓ PASS - 9 CLASSES
- FD3** Intra-protein consistency Identical WT FD_mr for hotspots of the same gene (same WT sequence) Identical WT FD_mr for the 4 hotspots of each Block A gene ✓ PASS - 5/5 GENES
- PS1** Correct classification of pathogenic variants The simulated index (0-1 scale, not the real PS_v4) classifies the panel mutations as Likely Pathogenic or Pathogenic ≥ 38/40 correctly classified ✓ PASS - 40/40
- PS2** Prioritized therapeutic targets Target mutations of approved drugs get a high simulated index (0-1 scale, not the real PS_v4, whose thresholds are 1.2 and 2.5) Simulated index > 0.6 in ≥ 5/6 therapeutic targets ✓ PASS - 6/6

Table: FULL PANEL - 40 WT/MUTANT PAIRS

#	Gene	Block	Type	Main cancer	Mutation	Molecular mechanism	Pathogenicity	Actions
21	BRAF	B	Proto-oncogene	Melanoma (50%), thyroid	V600E	Constitutive MAPK activation - vemurafenib target	Pathogenic (ClinVar)	Analyze 3D
22	BRAF	B	Proto-oncogene	Melanoma - resistance	V600K	Constitutive MAPK activation - variant	Pathogenic (ClinVar)	Analyze 3D
23	PIK3CA	B	Oncogene	Breast, endometrium, colon	H1047R	Constitutive PI3K activation - Akt/mTOR	Pathogenic (ClinVar)	Analyze 3D
24	PIK3CA	B	Oncogene	Breast, colon	E545K	Constitutive PI3K activation	Pathogenic (ClinVar)	Analyze 3D
25	PTEN	B	Tumor suppressor	Prostate, breast, endometrium	R130Q	Loss of phosphatase activity - PI3K hyperactivation	Pathogenic (ClinVar)	Analyze 3D
26	PTEN	B	Tumor suppressor	Cowden syndrome, multiple	C124S	Loss of catalytic phosphatase activity	Pathogenic (ClinVar)	Analyze 3D
27	APC	B	Tumor suppressor	Colon (FAP, sporadic)	R1450W	Loss of Wnt/β-catenin regulation	Pathogenic (ClinVar)	Analyze 3D
28	RB1	B	Tumor suppressor	Retinoblastoma, osteosarcoma	R661W	Loss of G1/S cell cycle regulation	Pathogenic (ClinVar)	Analyze 3D
29	CDKN2A	B	Tumor suppressor	Melanoma, pancreas, lung	R24P	Loss of CDK4/6 inhibition	Pathogenic (ClinVar)	Analyze 3D

Figure 28. Filters of the full 40-pair panel and per-row actions.

- 1 All (40):** shows all 40 pairs.
- 2 Block A (20):** only the Block A pairs.
- 3 Block B (20):** only the Block B pairs (the active filter in the screenshot).
- 4 Per-row actions:** “🔍 Analyze” loads that pair into the analysis section and “🏗 3D” opens its structure in the viewer. For pairs without a preloaded protein, “📄 Manual” takes the name and the mutation to the analysis form so you can paste the WT sequence by hand.

9.2 System results and export

32	ERBB2	B	Oncogene	HER2+ breast (20-30%)	V777L	Constitutive HER2 activation – trastuzumab target	Pathogenic (ClinVar)	Analyze	3D
33	MET	B	Proto-oncogene	Lung, renal, hepatic	Y1235D	Constitutive MET receptor activation	Pathogenic (ClinVar)	Analyze	3D
34	IDH1	B	Metabolic oncogene	Glioma, AML	R132H	Enzymatic neomorphism – 2-HG production	Pathogenic (ClinVar)	Analyze	3D
35	IDH2	B	Metabolic oncogene	AML, cholangiocarcinoma	R172K	Enzymatic neomorphism – 2-HG production	Pathogenic (ClinVar)	Analyze	3D
36	FGFR3	B	Proto-oncogene	Bladder, multiple myeloma	S249C	Constitutive FGFR3 activation – S-S bridge	Pathogenic (ClinVar)	Analyze	3D
37	ALK	B	Oncogene	Neuroblastoma, lung	F1174L	Constitutive ALK activation – crizotinib target	Pathogenic (ClinVar)	Analyze	3D
38	VHL	B	Tumor suppressor	Renal carcinoma, hemangioblastoma	Y98H	Loss of HIF-1 α regulation – pseudohypoxia	Pathogenic (ClinVar)	Analyze	3D
39	SMAD4	B	Tumor suppressor	Pancreas, colon	R361H	Loss of TGF- β /SMAD signaling	Pathogenic (ClinVar)	Analyze	3D
40	CTNNB1	B	Proto-oncogene	Colon, hepatic, endometrium	S45F	Constitutive Wnt/ β -catenin activation	Pathogenic (ClinVar)	Analyze	3D

SYSTEM RESULTS – PROTEINDELTA V4.0 PIPELINE

1 Export CSV

Simulation data, not pipeline data. The values in this table come from the application's deterministic simulation engine (constant VAL_RESULTS), not from an AlphaFold2/ColabFold or Biopython run. The classifications in the last column are hand-written and are not produced by classifyPath(). They must not be cited as experimental validation of the pipeline.

The "PS_v4" column in this table does NOT use the same scale as the real engine (classifyPath(), Analysis/Fractal section): this is a 0-1 scale with placeholder values, whereas classifyPath() produces an unbounded score where Pathogenic requires > 2.5. The values 0.68-0.996 shown here would not be classified as pathogenic if they were passed through classifyPath() – the classification in this column is not the result of applying any verifiable threshold; it is a manual label.

Structure of the 40 WT/Mutant pair panel and associated metrics: pLDDT, Co RMSD, ASASA, fractal dimension FD_bc and FD_mr, and PS_v4 pathogenicity score. **Reproducibility:** use the Analyze button in the pairs panel to load each gene into the analysis module and verify the results individually.

#	WT – MUT pair	Source	pLDDT WT	pLDDT MUT	RMSD (Å)	ASASA (Å ²)	FD_bc WT	FD_mr WT	Dir. AFD	PS_v4 (0-1)	Classification
1	TP53 R175H	REAL	94.8	93.7	0.717	-0.4	2.10	0.00	- neutral	0.901	PATHOGENIC
2	TP53 R248W	SIM	93.6	91.5	1.206	+16.5	1.00	1.00	- (sim, no real AFD)	0.993	PATHOGENIC
3	TP53 R248Q	SIM	93.6	92.8	0.291	+2.2	1.00	1.00	- (sim, no real AFD)	0.683	LIKELY PATHOGENIC
4	TP53 R273H	SIM	93.6	92.5	1.239	-2.0	1.00	1.00	- (sim, no real AFD)	0.980	PATHOGENIC
5	KRAS G12D	SIM	94.5	91.6	1.200	+6.4	1.00	1.00	- (sig. pos. real AFD)	0.990	PATHOGENIC

Figure 29. System results table and the export button.

1 Export CSV: downloads the results table.

Reading this table

The amber box above the table warns that these values come from the simulation engine and that the PS_v4 column uses a different scale from the real engine. The “Source” column tells apart pairs with real data (✓ REAL) from simulated ones (SIM).

Figure 30. *Detail of the system results table.*

10. Breast Validation v5.0

A standalone module for human breast cancer (GRCh38/hg38). It reproduces the panel of 40 WT vs. mutant pairs and adds 12 benign controls. The amber box at the top states which results are simulated and which come from real runs.

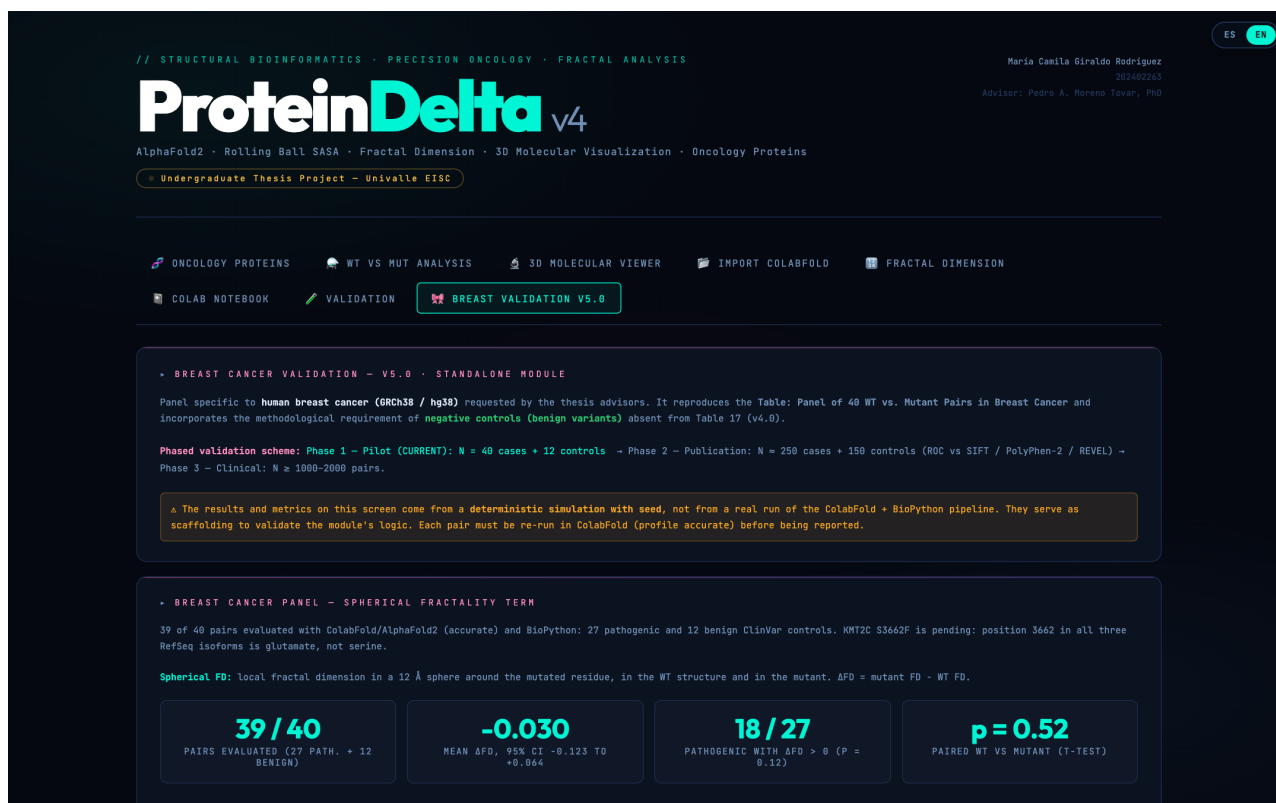


Figure 31. Top of the Breast Validation module.

10.1 Spherical fractality term

This block holds real ColabFold/AlphaFold2 and BioPython data. The spherical FD is the local fractal dimension in a 12 Å sphere around the mutated residue. The cards summarize the pairs evaluated, the mean Δ FD with its 95 % CI, the pathogenic variants with Δ FD > 0 and the paired t-test; the table shows, for each pair, the WT and mutant FD, Δ FD (with a bar on a ± 1 scale), the change in PS_v4 and the resulting class.

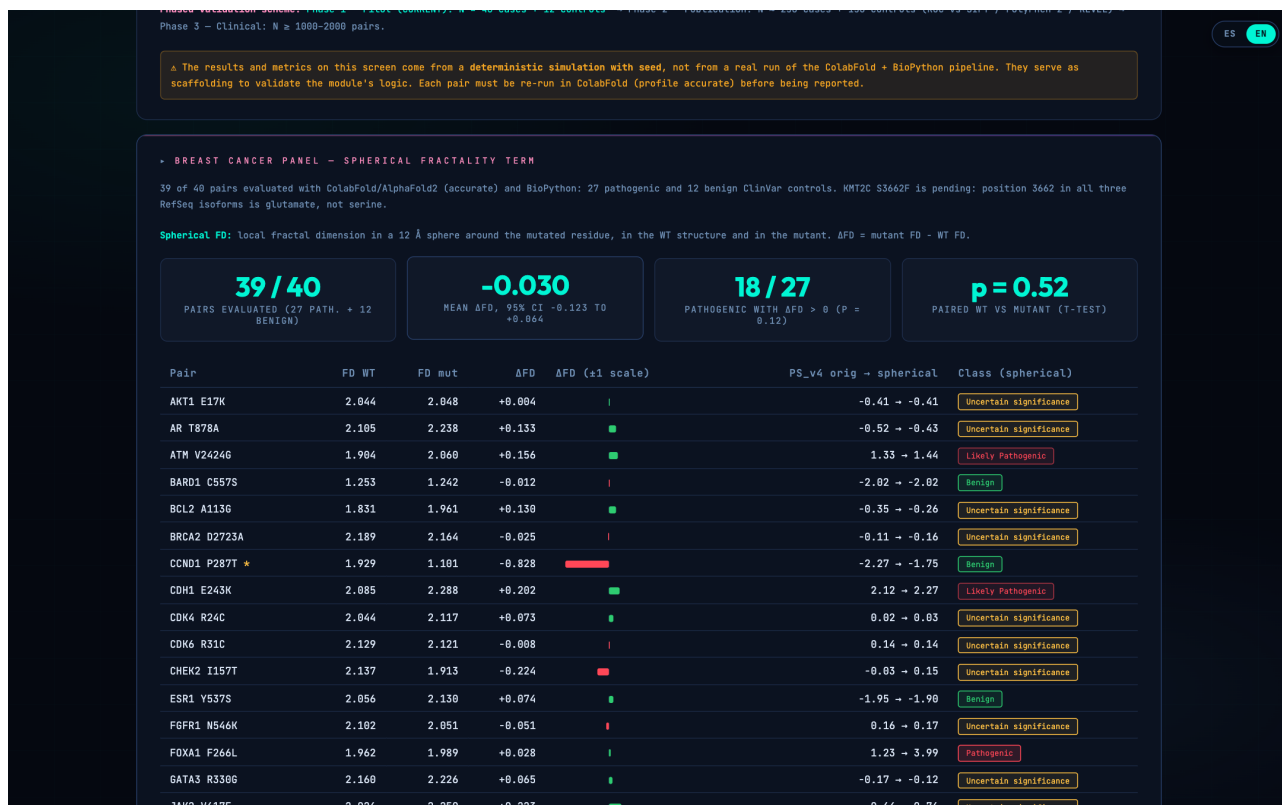


Figure 32. Spherical fractality term panel.

10.2 Cases: filters and actions

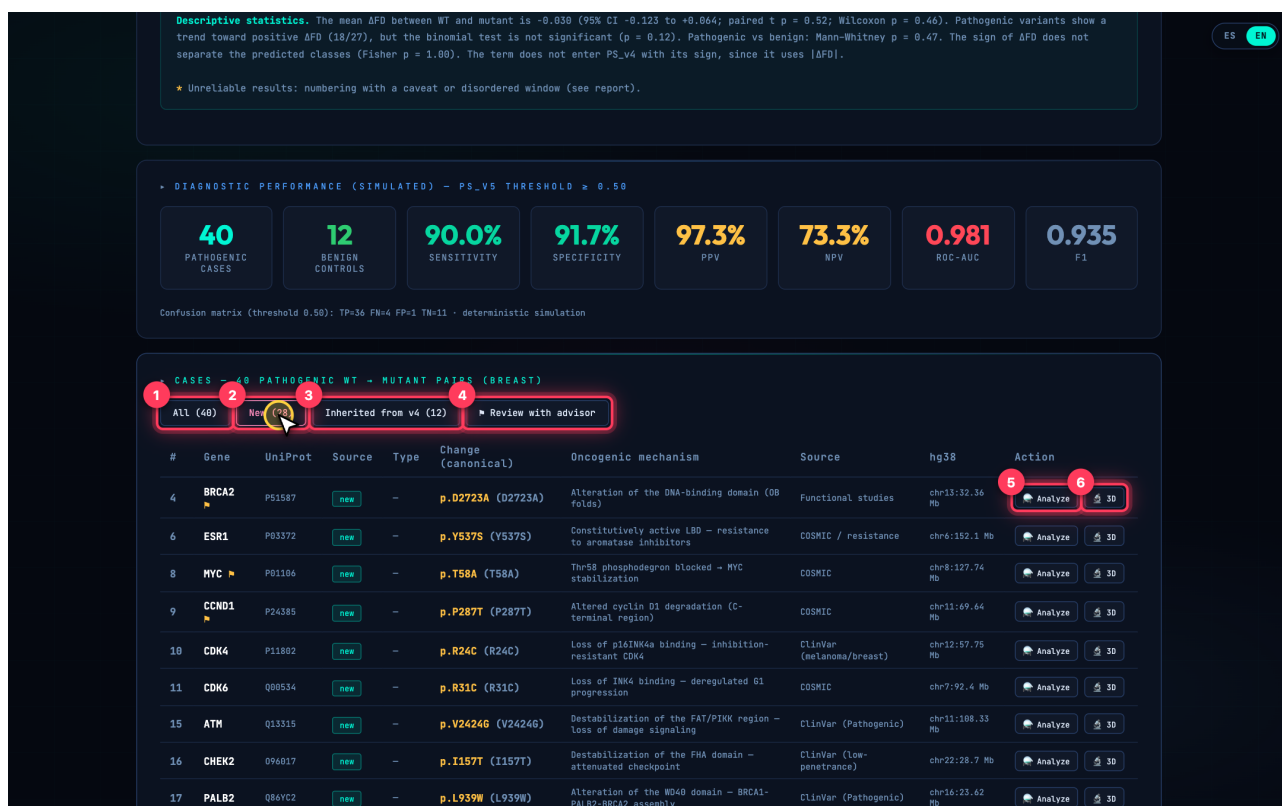


Figure 33. Case table filters and per-row actions.

- 1 All (40)
- 2 New (28): pairs added in v5 (the active filter in the screenshot).
- 3 Inherited from v4 (12): pairs carried over from the previous validation.

- 4  **Review with advisor:** pairs flagged for review.
- 5  **Analyze:** takes the pair to the analysis section.
- 6  **3D:** opens the structure in the viewer.

Clicking “ Analyze” switches to “WT vs MUT Analysis” with the sequence, the name and the mutation already loaded; all that is left is to click “ Analyze WT vs Mutant”.

SEQUENCE SETUP

WT (WILD-TYPE) SEQUENCE – AUTO-FILLED WHEN UPLOADING PDB IN "IMPORT COLABFOLD"

CEEELDLITSLQNRDIQMRIKKKQRQRVFPQPSLYLAKTSTLPRISLKAAVGGQVPSACSHKQLY
TYGVSKHKIKINSKNAESQFHTEDYFGKESLWTGKGIQADGGWLPSNDGKAGKEEFYRALCDTPGV
DPKLISRIWVYNYHRIIWKLAAMECAFPKEFANRCLSPERVLLQLKYRYDTEIDRSRRSAIKKIMERO
DTAAKTLVLCVSDIISLANSISETSSNKTSSADTQKVAIIEITDGVYAVKAQLDPPLLAVLKNGRLTVG

NAME / PROTEIN
BRCA2_D2723A_mama

MUTATION MODE
Point mutation list

ADD MUTATION (E.G. D10L)

D10L

D251A

The analysis includes **Fractal Dimension** Box-Counting (FD_bc) and Mass-Radius (FD_mr)
– global, per 21-residue sequence window, and locally in the 3D microenvironment of
the mutated residue (r=10Å). AFD is integrated into the PS pathogenicity score.

Analyze WT vs Mutant

Figure 34. Pair BRCA2 D2723A loaded into the analysis form from the breast module.

10.3 Export

31	case	PDXA1 P288L	91.7	98.7	1.058	+19.5	0.136	0.905	PATHOGENIC	✓
32	case	KMT2C S3662F	88.5	86.8	0.964	+5.0	0.079	0.828	LIKELY PATHOGENIC	✓
33	case	SF3B1 K708E	92.4	91.2	1.574	+10.8	0.039	0.826	LIKELY PATHOGENIC	✓
34	case	BCL2 A113G	96.9	95.4	1.288	+1.4	0.047	0.632	LIKELY PATHOGENIC	✓
35	case	RAD51D C119W	91.3	90.8	1.519	+16.2	0.069	0.851	PATHOGENIC	✓
36	case	BARD1 C557S	94.5	94.0	1.646	+7.4	0.088	0.815	LIKELY PATHOGENIC	✓
37	case	STK11 T189M	95.2	92.4	0.926	+16.9	0.089	0.814	LIKELY PATHOGENIC	✓
38	case	JAK2 V617F	91.6	89.0	0.396	+7.3	0.012	0.510	LIKELY PATHOGENIC	✓
39	case	STAT3 D661Y	96.7	94.9	1.164	+8.1	0.040	0.691	LIKELY PATHOGENIC	✓
40	case	VEGFA P114L	89.1	86.7	0.928	+1.9	0.038	0.630	LIKELY PATHOGENIC	✓
41	control	TP53 P72R	98.6	88.2	0.371	-2.7	0.008	0.131	BENIGN	✓
42	control	BRCA1 S1613G	93.4	91.4	0.201	+0.8	0.026	0.084	BENIGN	✓
43	control	BRCA1 P871L	92.4	91.3	0.362	+0.6	0.027	0.138	BENIGN	✓
44	control	BRCA2 N372H	89.1	87.0	0.361	+3.2	0.035	0.529	LIKELY PATHOGENIC	✗
45	control	BRCA2 V2466A	94.8	92.6	0.427	-2.2	0.013	0.068	BENIGN	✓
46	control	ATM D1853N	98.1	89.2	0.294	+2.9	0.023	0.154	BENIGN	✓
47	control	PALB2 Q559R	94.7	94.2	0.173	+0.2	0.025	0.055	BENIGN	✓
48	control	ERBB2 I655V	88.7	88.6	0.558	+3.0	0.012	0.478	LIKELY BENIGN	✓
49	control	CDH1 A592T	89.8	88.3	0.231	0.0	0.002	0.132	BENIGN	✓
50	control	BARD1 V507M	89.6	89.4	0.243	+0.8	0.011	0.107	BENIGN	✓
51	control	STK11 F394L	98.3	87.9	0.335	+1.3	0.020	0.162	BENIGN	✓
52	control	MET N375S	94.5	93.6	0.400	-3.6	0.017	0.043	BENIGN	✓

1
↓ Export CSV (52 rows)

Figure 35. Results table (cases + controls) and the export button.

- 1 Export CSV (52 rows):** downloads the 40 cases and the 12 controls.

11. Changing the language

The ES / EN selector is fixed in the top-right corner and visible from every section. The choice is remembered for your next visit. This manual uses the English interface.

Step 1. Click “ES” to switch the interface to Spanish (or “EN” to return to English).



Figure 36. Clicking ES in the language selector.



Figure 37. Selector with ES active: the header is now in Spanish.

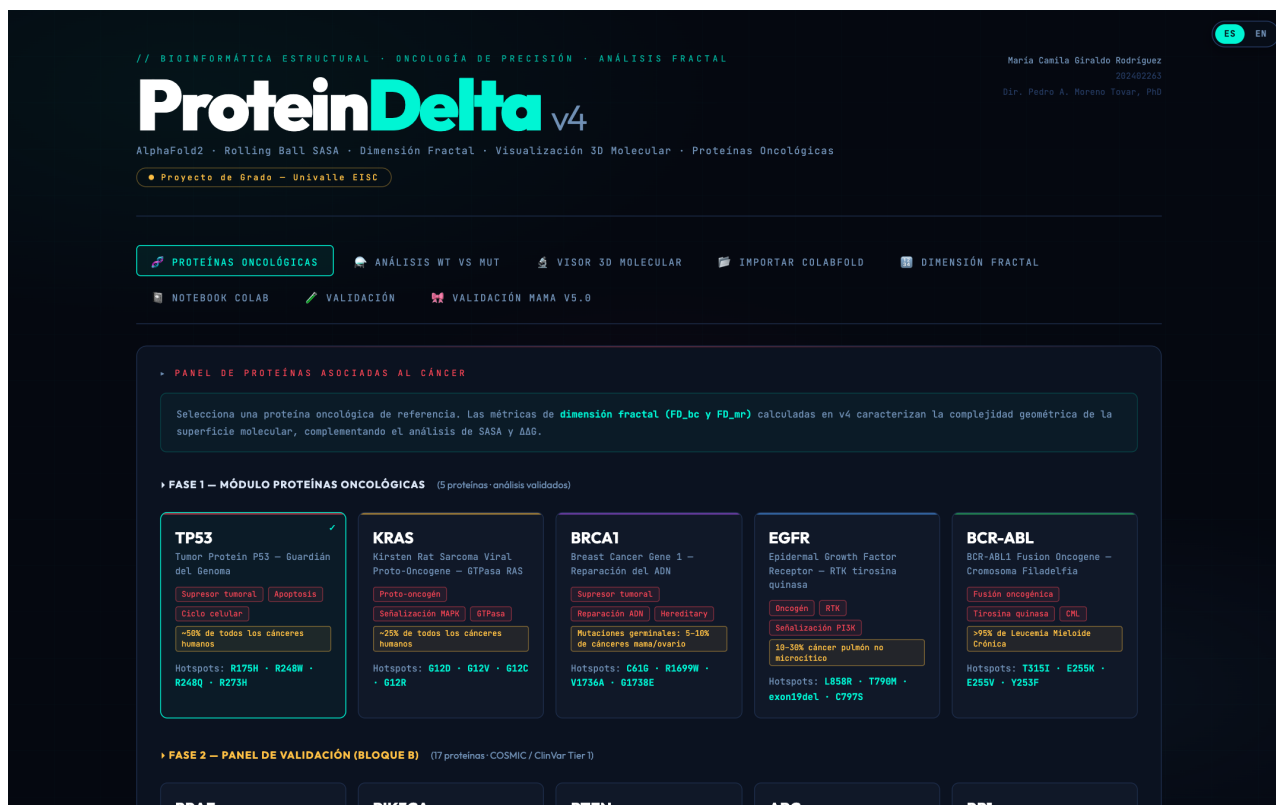


Figure 38. The whole application in Spanish.

The translation also applies to content the application generates later (tables, metrics, warnings). Numbers and gene and mutation names do not change.

12. Recommended workflows

12.1 Quick exploration (simulated mode)

- Step 1.**  Oncology Proteins → pick a protein → pick a mutation → “Load into Analysis”.
- Step 2.**  WT vs MUT Analysis → “⚡ Analyze WT vs Mutant”.
- Step 3.** Review metrics, classification and tabs; look at the structure in the  3D Molecular Viewer.
- Useful for getting to know the tool and its logic. The values must not be cited as results.*

12.2 Real-data analysis

- Step 1.**  Colab Notebook → download the notebook.
- Step 2.** Run it in Google Colab with a GPU and the accurate profile; download all_outputs.zip.
- Step 3.**  Import ColabFold → load the JSON files, both CSV files and, if possible, the PDB files → “Analyze with real data”.
- Step 4.** Review the results in  Analysis and  Fractal Dimension; look at the models in the  3D viewer and highlight the microenvironment.
- Step 5.** Export full_analysis.csv, results.json or all_outputs.zip.

13. Troubleshooting

Situation	What to do
The 3D viewer stays empty.	Check your internet connection (structures are downloaded from RCSB) and that WebGL is enabled in the browser. Try “Reset” or “Center”.
“Analyze with real data” shows a warning and does not run.	rmsd_per_residue.csv or delta_sasa.csv is missing. Load them from all_outputs.zip.
The local spatial FD shows “Not available”.	It needs the real WT and mutant PDB files in “Import ColabFold”. It does not exist in simulated mode.
FD_mr shows “not computable”.	The Mass-Radius fit was not valid for that structure; the application does not report an unreliable value.
The mutation is not accepted.	Use original residue + position + new residue with one-letter uppercase codes (e.g. R175H).
The context bar says “simulated”.	The analysis was run without ColabFold files. For real data, follow workflow 12.2.

14. Glossary

Term	Meaning
WT / MUT	Wild-type protein / protein carrying the mutation.
pLDDT	AlphaFold2 per-residue confidence (0–100). Higher = more reliable.
$\Delta\Delta G$	Change in folding free energy caused by the mutation (kcal/mol). Positive = destabilizing.
Evolutionary fitness	How tolerant a position is to change, based on evolutionary conservation (0–1).
C α RMSD	Displacement of each residue's alpha carbon between WT and mutant after superposition (Å).
$\Delta SASA$	Change in solvent-accessible surface area (Å ²).
FD_bc / FD_mr	Fractal dimension by Box-Counting / Mass-Radius.
Spherical FD	Local FD in a 12 Å sphere around the mutated residue.
PS_v4	Pathogenicity score combining $\Delta\Delta G$, fitness, $\Delta pLDDT$ and ΔFD_{bc} .
VUS	Variant of uncertain significance.
PDB	File format and database of 3D protein structures (RCSB Protein Data Bank).
ColabFold	An implementation of AlphaFold2 that runs in Google Colab.